What the occupation law of a finite Markov chain determines: principal minors, modal supports and cycle trees

Arjun Pemmasani*

Department of Mathematics, Harvey Mudd College, 301 Platt Boulevard, Claremont, 91711, CA, USA

Abstract

For an irreducible generator Q on d states, the Donsker and Varadhan functional of the occupation measure determines exactly the principal minors of $-Q$. This answers the question raised by the 3-cycle example of Kim and Landim (Stochastic Process. Appl. 189 (2025) 104733). The generators with these minors form a finite set. It consists of Q and its time reversal when $-Q$ has no rank-one cut, and of an orbit of cut-transposes in general. From the stationary start no statistic of the occupation counts separates Q from its reversal. For a chain that switches about $\tau \log N$ times in N steps, the support of the most likely vector of counts under killing at known rates retains the same information. For a reversible chain on a known graph with m edges a design fixed in advance reads $m + d - 1$ boundary coordinates, which is minimal. It recovers Q stably, with a finite-horizon correction uniform over a compact class. At low temperature, with Arrhenius rates, the unperturbed modal path misses cycles, and killing the complement of a face masks its depth. A reward of known size on the face does neither. The largest reward exponent at which a cycle still wins is its depth. The depth of every face is the largest such threshold over its subfaces, and a set is a cycle exactly when every proper subset is strictly shallower. So the modal supports determine the Freidlin and Wentzell cycle tree, without reversibility or nondegeneracy.

Keywords: identifiability, principal minors, occupation measure, large deviations, Perron root, metastability, Freidlin–Wentzell cycles
2020 MSC: 60J27, 60F10, 60J10, 15A15, 15A18, 15A29, 15B48

1. Introduction

Let Q be an irreducible generator on $[d] = \{1, \dots, d\}$, and let J_V be the Donsker and Varadhan functional of its occupation measure [11]. We prove that J_V determines exactly the principal minors of $-Q$ (Theorem A). The generators with these minors form a finite set. It consists of Q and its time reversal Q^* when $-Q$ has no rank-one cut, and it is an orbit of cut-transposes in general. Kim and Landim [18, Section 5] show that J_V determines the minors of orders 1 and 2, and they observe that the uniform 3-cycle and its reversal share J_V. Theorem A says exactly which generators share it.

*Corresponding author.
Email address: apemmasani@hmc.edu (Arjun Pemmasani)

The functional enters through the occupation counts of a chain that rarely switches. Fix a start law and run the chain for N steps with transition matrix $I+(T/N)Q$. With $T=\tau\log N$ the chain makes about $\tau\log N$ switches. Its composition is the vector of occupation counts, and the support of a composition is a face of the simplex. In [22] we describe the law of the composition on every face at this scale. On a face F the most likely composition has probability $N^{-\Phi_\tau(F)+o(1)}$, with the score $\Phi_\tau(F)=|F|-1+\tau\lambda_F$ and λ_F the principal exit rate of F. One power of N per dimension competes with the cost of staying in the face, and the mode moves to larger faces as τ grows. Inside a face the logarithm of the law is T times J_V.

We ask which coarse functionals of the law of the counts determine Q, and what each of them loses. We compare 3, from fine to coarse. The first is the law itself, at every N and T. The second is its logarithmic scale on single compositions, the functional J_V. The third is a finite label, the support of the most likely composition when the chain is killed at known rates. The third determines exactly the same minors as the second (Theorem B). We then read a reversible generator from finitely many boundaries of a design fixed in advance (Theorem C), and the Freidlin and Wentzell cycle tree of an Arrhenius family from the same labels (Theorem D).

1.1. The Donsker and Varadhan functional

Write $Q^*=D_\pi^{-1}Q^\top D_\pi$ for the time reversal of an irreducible generator with stationary law π. Call a partition $[d]=X\cup Y$ with $|X|,|Y|\geq 2$ a cut of Q if both blocks $Q[X,Y]$ and $Q[Y,X]$ have rank at most 1. Across a cut a cut-transpose replaces the 2 rank-one blocks by the products of their factors taken crosswise and transposes the block on Y (Section 3.3).

Theorem A (what the Donsker and Varadhan functional determines)**.** *Let Q and Q' be irreducible generators on $[d]$.*

(i) $J_V=J'_V$ on the simplex if and only if $-Q$ and $-Q'$ have the same principal minors.

(ii) The irreducible generators with the principal minors of $-Q$ form a finite set. It contains Q and Q^*. It is $\{Q,Q^*\}$ when Q has no cut, in particular when $d\leq 3$, and $\{Q\}$ when Q is reversible. In general it is the set of generator points of the cut-transpose orbit of $-Q$, and it can contain generators other than Q and Q^*.

(iii) From the stationary start the law of the composition is the same under Q and under Q^*, for every N and every T with $Tq_{\max}\leq N$.

Part (i) is Theorem 3.3, part (ii) is Theorem 3.5 with the example after it, and part (iii) is Proposition 3.7. Part (i) has 3 inputs. The first is the variational formula $\lambda_V(v)=\min_\alpha(J_V(\alpha)-v\cdot\alpha)$ for $\lambda_V(v)=-\varrho(Q+\operatorname{diag}v)$. The second is the irreducibility of the multiaffine polynomial $\det(-Q_F-\operatorname{diag}w)$ for every irreducible face, which is Lemma 5.1 of Al Ahmadieh [1]. The third is the Perron-branch lemma (Lemma 3.2). It says that the Perron root on an open set of fields determines that polynomial, hence every principal minor. Kim and Landim use the minors of orders 1 and 2, with the stationary law, to fix a reversible generator. By their Corollary 5.8 the joint functional of the empirical measure and the empirical flow determines the generator, so adding the flow removes the whole fiber. Part (iii) shows that the fiber is not an artefact of the logarithmic scale. From the stationary start even the exact law of the counts cannot separate Q from Q^*. For a start outside a finite

union of proper subspaces the second-order amplitudes of the local law do separate every member of the fiber (Proposition 3.8).

1.2. Modal supports

Weight each composition by $e^{Tv\cdot n/N}$ for a field $v \le 0$. This is the chance that the chain survives when it is killed at state i at the known rate $-v_i$. We record only which face carries the most likely surviving composition, as a function of (τ, v), and call this the modal-support map. The data are the boundaries between the open regions where one face wins strictly (Definition 4.3).

Theorem B (what the modal supports determine)**.** *Let Q and Q' be irreducible generators and let the start have full support. They have the same modal-support map, outside a closed nowhere dense set of (τ, v), if and only if $-Q$ and $-Q'$ have the same principal minors. If the minors differ, there is a nonempty open set of (τ, v) on which both maps have a strict winner and the 2 winners differ.*

This is Theorem 4.7, with Proposition 4.6 for the last sentence. The proof is a descent through the faces. For every irreducible face F and a generic field on F, the exposure lemma (Lemma 4.5) finds a proper subface G that shares a boundary of strict regions with F. Along that boundary $\lambda_F = \lambda_G - (|F| - |G|)/\tau$. Induction on $|F|$ then traces λ_F on an open set of fields, and the Perron-branch lemma extends it to all fields.

Theorem B is a statement about retention, not about sampling. The survival probability under calibrated killing is a finer observation. It equals $e^{-T\lambda_V(v)}$ up to a bounded factor, so it gives every principal minor at once, with no descent (Remark 3.4). The modal support is not easier to sample. The most likely composition carries probability $N^{-\Phi+o(1)}$, so estimating which face carries it needs far more independent runs than estimating the survival probability. We know of no experiment in which the modal support is observed and the survival probability is not. But the forward theory of [22] outputs the support of the most likely composition, and so does any exact optimizer that reports only a maximizer of the weighted law. Theorem B says that this output loses nothing that J_V keeps.

What is borrowed and what is new. The principal minors are not observed. Theorem 3.3 reads all of them off J_V, and Theorem 4.7 reads all of them off the modal-support map, through the exposure lemma and the Perron-branch lemma. This is our main contribution. The fiber of the principal-minor map of an irreducible matrix is known, up to diagonal similarity and transposition, from Loewy [20], Hartfiel and Loewy [16], Al Ahmadieh [1] and Chatterjee, Ghosh, Gurjar and Raj [7], who describe it by sequences of cut-transposes (Section 3.3). Our second contribution, and the shortest step, translates this fiber to generators. A Perron and Frobenius normalization picks exactly one generator in each class of diagonal similarity. So the generator fiber is finite, every member is irreducible with the spectrum of Q, and a cut-transpose acts as Q on one side of the cut and as Q^* on the other (Theorem 3.5 and Proposition 3.8). The third is the stationary obstruction of part (iii) of Theorem A.

1.3. A reversible design and finite horizons

For a reversible chain on a known connected graph with m edges the fields can be fixed before anything is observed.

Theorem C (a reversible design)**.** *Let Q be reversible and irreducible on a known connected graph with m edges, and suppose that known constants R and b bound the norm of the symmetrized generator above and every edge product $\sqrt{q_{ij}q_{ji}}$ below.*

(i) A design whose fields and time scales depend only on R, b and the labels reads $d-1$ singleton switches and m pair-entry times off the modal-support map, and these $m+d-1$ numbers determine Q.

(ii) No family of fewer than $m+d-1$ scalar C^1 observations, with finitely many locally constant labels, identifies Q on any open set of such generators.

(iii) The reconstruction is Lipschitz in the observed coordinates, with constants that depend only on d and b/R.

(iv) At the finite horizon N, with $L=\log N$, the observed pair-entry times are $t_{ij}(1-\log L/(2L))+c_{ij}/L+O(\log L/L^2)$, uniformly over the generators of the class and the starts with $\min_i \mu_i \geq \mu_*$, and the start bound cannot be dropped.

Parts (i) to (iii) are Theorems 5.2, 5.3 and 5.5, and part (iv) is Theorem 5.7 with Remark 5.8. The coordinates are those of Wolfer and Watanabe [25], and Kim and Landim [18, Lemma 5.1 and Corollary 5.3] recover the same quantities from J_V. Theorem C adds a design fixed in advance, its minimality, its stability with constants from R and b alone, and its finite-horizon behaviour.

1.4. Low temperature and the cycle tree

Let the rates be $q_{ij}=e^{-V_{ij}/\eta}$ with fixed costs $V_{ij}\geq 0$ and a temperature $\eta\downarrow 0$, and let the time scale be $\tau=e^{h/\eta}$ for a height h. Then dimension and exit rate compete on the Freidlin and Wentzell cycles of the chain. The depth $D(F)$ of a face is the exponential order of its exit rate, and it is the largest depth of a cycle inside F (Theorem 6.1). Away from finitely many heights the winner is a cycle of least size among those deeper than h (Theorem 6.2). At a critical height explicit prefactors decide, and a noncycle can win on an open window of the rescaled time scale (Theorem 6.4 and Example 6.5).

So modal winners need not be cycles. Do the modal supports determine the cycles at all? The 2 obvious experiments do not. The unperturbed modal path shows the depth of each face it selects and nothing about the cycles that are never the smallest eligible face. Killing the complement of a face at a large rate does not show its depth either. The whole space then has an exit rate of the same order, and the winner leaves the face only at a larger height. The cure is to reward the face instead. For a proper face G, a constant $c>0$ and a real γ we put the field $v^{G,\gamma}=c\,e^{-\gamma/\eta}\mathbf{1}_G$. The threshold $\vartheta(G)$ is the supremum of the γ for which G wins under $v^{G,\gamma}$ at some height above γ.

Theorem D (the cycle tree from calibrated rewards)**.** *Let (Q_η) be an Arrhenius family with all rates positive, and let the start have full support.*

(i) $\vartheta(G)\leq D(G)$ for every proper face G, with equality when G is a cycle.

(ii) $D(F)=\max_{G\subseteq F}\vartheta(G)$ for every proper face F.

(iii) A nonempty proper set C is a cycle if and only if $D(C') < D(C)$ for every nonempty proper subset C' of C.

So the modal-support map, restricted to the fields $v^{G,\gamma}$, determines the depth of every face, the cycles, their depths and the cycle tree. No step uses reversibility, and no nondegeneracy of the depths is assumed.

Parts (i) and (ii) are Theorem 6.11 and part (iii) is Theorem 6.12. The design needs 3 calibrated quantities, the time unit, the amplitude of the reward and the labels of the states, and no rate of Q. The reward keeps a face whose rate is known, namely the whole space with $\lambda_V(v) \le 0$, and the killing design destroys exactly this. Criterion (iii) holds with ties between the depths of different faces, because Catoni's cycles are strict and depths increase strictly along nested cycles (Lemma 6.10). Its weak form, with $D(C') \le D(C)$, fails already on 3 states. Finally the endpoints of the critical windows determine the prefactors of the faces on the lower envelope, and only bound the others (Theorem 6.13).

1.5. Related work

The rate inside a face is the Donsker and Varadhan functional of the killed chain [11], and the exact law of the counts goes back to Good [15]. Bertini, Gabrielli and Landim [5] and Landim [19] expand the level-two rate functions of finite chains in a parameter, and Kim and Landim [18] treat the joint rate of the empirical measure and the flow [3, 4]. Their Section 5 is the starting point of Theorem A. The matrix theory behind the fiber is credited above and in Section 3.3, together with Engel and Schneider [12]. Cohen [9] and Friedland [14] study the convexity of $v \mapsto \varrho(A + \operatorname{diag} v)$, and Delmas, Dronnier and Zitt [10] prove a multiplicative analogue of the Perron-branch lemma. The cycle framework is that of Freidlin and Wentzell [13], Catoni [6], Olivieri and Vares [21], Trouvé [23] and Cirillo, Nardi and Sohier [8]. The eigenvalue asymptotics behind the prefactors go back to Wentzell [24]. New at low temperature are the score of dimension against exit rate, the competition of envelopes at a critical depth with its noncycle window, and the reading of the cycle tree from calibrated rewards.

1.6. Plan of the paper

Section 2 fixes the notation and states the results of [22] that we use. They are the first-order results, the face maxima with a field, their uniform version on a compact class and the constant of the local law. Section 3 proves Theorem A. Section 4 sets up the modal-support map and proves Theorem B. Section 5 treats the reversible design and the finite horizon, and Section 6 the low temperature. Section 7 lists the computations that check the paper and the open questions.

2. Setting and imported results

We fix the notation and state, without proof, the results of [22] that we use, each with a pointer to its place there.

2.1. Setting

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We write q for the vector (q_i). The start μ is a probability law on $[d]$, and $V = [d]$ is the face of all states. For $N \geq 2$ and $T > 0$ put $L = \log N$ and $\tau = T/L$. The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + (T/N)Q$, which is stochastic when $Tq_{\max} \leq N$. The composition is the vector n of visit counts, $n_i = \#\{t \leq N : X_t = i\}$, so $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for its law, $\operatorname{supp} n = \{i : n_i > 0\}$ and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. The killed generator of F is $Q_F = (Q_{ij})_{i,j \in F}$. Its diagonal keeps the full exit rates q_i, so it generates the chain killed when it leaves F. For $A \subseteq F$ the principal submatrix of $-Q_F$ on A is $-Q_A$. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$. We call F irreducible if G_F is strongly connected, so every singleton is irreducible. We call F admissible if some path of positive probability has support exactly F, and we say that F satisfies (H) if it is irreducible and $\mu(F) > 0$.

Perron roots and fields. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and its Perron root $\varrho(B)$ is then its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [2]. For a face F and a field $v \in \mathbb{R}^d$ put

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v_F\big), \qquad \Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v),$$

and $\lambda_F = \lambda_F(0)$, $\Phi_\tau = \Phi_{\tau,0}$. The number λ_F is the exit rate of F, and $\lambda_{\{i\}}(v) = q_i - v_i$. If F is reducible, Q_F is block triangular in an order of its strongly connected components, and $\lambda_F(v)$ is the least of the numbers $\lambda_C(v)$ over these components C. The shift rule $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$ holds for every face. If F is irreducible, let r and l be positive right and left Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$, and put $\alpha_F^*(v) = l \circ r$, the quasi-ergodic law of F under the field, where $x \circ y$ is the entrywise product. Finally

$$M_F(T, v) = \max\big\{P_N(n)\, e^{Tv \cdot n/N} : \operatorname{supp} n = F\big\}, \qquad \max \emptyset = 0.$$

The functional. For $\alpha \in \Delta_F$ let

$$J_F(\alpha) = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(-\frac{(Q_F u)_i}{u_i}\Big),$$

the Donsker and Varadhan functional of the chain killed on leaving F [11], and J_V that of the whole chain.

2.2. Imported results

Lemma 2.1 (admissible faces, [22, Lemma 2.2(b), (c) and Theorem 4.2(ii)])**.** (a) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces, and if μ has full support every irreducible face is admissible.* (b) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.* (c) *A face that is not admissible carries no composition of positive probability.*

Lemma 2.2 (monotonicity in the face, [22, Lemma 3.2]). *If $F \subsetneq G$ are faces and G is irreducible, then $\lambda_G(v) < \lambda_F(v)$ for every field v. In particular every proper face of an irreducible chain has $\lambda_F > \lambda_V = 0$.*

Proposition 2.3 (the variational formula, [22, Propositions 3.1 and 3.4]). *Let F be a face.*

(a) $J_F = J_V$ on Δ_F.

(b) If F is irreducible, then for every $v \in \mathbb{R}^F$

$$\lambda_F(v) = \min_{\alpha \in \Delta_F} \big(J_F(\alpha) - v \cdot \alpha \big),$$

with the minimum attained only at $\alpha_F^*(v)$. So $-\lambda_F$ is the Legendre transform of J_F, and J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°. The map $v \mapsto \lambda_F(v)$ is concave and real-analytic, with $\partial \lambda_F / \partial v_i = -\alpha_F^*(v)_i$.

(c) If F is irreducible, Q_F is reversible with respect to some $\nu \in (0,\infty)^F$ and $S_F = D_{\sqrt{\nu}}(-Q_F)D_{\sqrt{\nu}}^{-1}$, then $\lambda_F(v) = \lambda_{\min}(S_F - \operatorname{diag} v_F)$.

The derivative in (b) is the first-order perturbation $\partial \varrho / \partial v_i = l_i r_i$ of a simple eigenvalue, with $l \cdot r = 1$.

Lemma 2.4 (a uniform upper bound, [22, Lemma 4.1 and (6)]). *Let F be irreducible with $k = |F| \geq 2$ and let $\mathcal{V} \subset \mathbb{R}^d$ be compact. There is C, depending on Q, μ and $\mathcal{V}$, such that for all $v \in \mathcal{V}$, all $N \geq 2$ and $T > 0$ with $T q_{\max} \leq N/2$, and every n with $\operatorname{supp} n = F$,*

$$P_N(n)\, e^{Tv \cdot n/N} \leq C \Big(\frac{T}{N} \Big)^{k-1} e^{-T\lambda_F(v) + CT^2/N}. \tag{1}$$

The bound comes from the Perron tilt $\theta = q - v - \lambda_F(v)\mathbf{1}$, and it holds for a path of any length in place of N, hence for each block of a path.

For a singleton the only composition is Ne_i, so by [22, (1)]

$$M_{\{i\}}(T, v) = \mu_i \Big(1 - \frac{T q_i}{N} \Big)^{N-1} e^{Tv_i} = \mu_i e^{-T(q_i - v_i)} \big(1 + O(T^2/N) \big). \tag{2}$$

Theorem 2.5 (first-order selection with a field, [22, Theorem 4.2 and Remark 4.3]). *Fix Q, μ and a field v.*

(a) Let F be admissible with $k = |F|$. There are κ, T_0 and N_0 such that $\log M_F(T, v) \leq -(k-1)L + (k-1)\log T - T\lambda_F(v) + \kappa$ for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$. If F is not admissible, then $M_F(T, v) = 0$.

(b) Let $T = \tau L$ with $\tau > 0$ fixed, and let $v = 0$ or let μ have full support. Then $\log \max_n P_N(n) e^{Tv \cdot n/N} = -L \min_F \Phi_{\tau, v}(F) + O(\log L)$ over the admissible faces, and for large N the support of every maximizer minimizes $\Phi_{\tau,v}$ among them.

Part (a) is the upper bound of [22, Theorem 4.2(ii)] with a field, and (b) is [22, Theorem 4.2(iii)] and its field form. We use (b) only at a fixed τ.

Definition 2.6 (tilt and constant, [22, Definition 5.1]). Let F be irreducible with $k = |F| \geq 2$. For $\alpha \in \Delta_F^\circ$ the supremum defining $J_F(\alpha)$ is attained at a vector r_α, unique up to a factor. Put $l_\alpha = \alpha / r_\alpha$ entrywise, and let $\widehat{Q}_\alpha$ be the generator on F with off-diagonal entries $q_{ij} r_{\alpha,j} / r_{\alpha,i}$, whose stationary law is α. With $D = \operatorname{diag} \alpha$ let $Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top$ and $\Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D$, and let Σ_α be Γ_α with the row and column of one state deleted. Then

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2} \sqrt{\det \Sigma_\alpha}}, \qquad K_F(v) = C_F\big(\alpha_F^*(v)\big),$$

and $K_{\{i\}}(v) = \mu_i$.

Proposition 2.7 (the constant, [22, Proposition 5.2 and the text after Definition 5.1]). *Let F be irreducible with $k \geq 2$. The maps $\alpha \mapsto r_\alpha$ (normalized), l_α, Σ_α, $J_F(\alpha)$ and $C_F(\alpha)$ are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 J_F(\alpha) = \Sigma_\alpha^{-1}$ in the coordinates $(\alpha_2, \dots, \alpha_k)$. At $\alpha = \alpha_F^*(v)$ the vector r_α is a multiple of the right Perron vector of $Q_F + \operatorname{diag} v_F$, and l_α is the matching multiple of the left one.*

Theorem 2.8 (face maxima with a field, [22, Theorem 5.7]). *Let F satisfy* (H) *with $k = |F|$, and let $\mathcal{V} \subset \mathbb{R}^d$ be compact. Uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,*

$$\log M_F(T, v) = -(k-1)L + \tfrac{k-1}{2} \log T + \log K_F(v) - T\lambda_F(v) + O_{\mathcal{V}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \tag{3}$$

and for $k \geq 2$ every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^*(v) + O_{\mathcal{V}}(1/T)$.

Theorem 2.9 (the stratified local law, [22, Theorem 5.6]). *For every face F satisfying* (H) *with $k = |F| \geq 2$ and every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/3}$,*

$$-\log P_N(n) = (k-1)L + T\, J_V(n/N) - \frac{k-1}{2} \log T - \log C_F(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

By Theorem 2.9, J_V is the logarithmic scale of the law on single compositions. Section 5.2 also imports the uniform version of Theorem 2.8 on a compact class of reversible chains (Proposition 5.6). It is stated there because it needs the design of Section 5. Nothing else from [22] enters.

3. The Donsker and Varadhan functional and the principal minors

This section proves Theorem A. In Section 3.1 the Perron root of a face on an open set of fields determines the principal minors of the face. Section 3.2 deduces that J_V determines exactly the principal minors of $-Q$. Section 3.3 describes the generators with given minors, and Section 3.4 what the exact law adds.

3.1. The polynomial of a face and the Perron branch

The principal minors of $-Q_F$ are the numbers $\det(-Q_A)$ for $A \subseteq F$, with $\det(-Q_\emptyset) = 1$. They are the coefficients of the multiaffine polynomial

$$R_F(w) = \det\bigl(-Q_F - \operatorname{diag} w\bigr) = \sum_{A \subseteq F} \det(-Q_A) \prod_{i \in F \setminus A} (-w_i), \qquad w \in \mathbb{R}^F.$$

For every field v, $\lambda_F(v)$ is a real root of $t \mapsto R_F(v + t\mathbf{1})$, and if F is irreducible it is the smallest real root, since the other real eigenvalues of $-Q_F - \operatorname{diag} v_F$ exceed the Perron eigenvalue.

Lemma 3.1 (irreducibility)**.** *If F is irreducible, then R_F is irreducible over $\mathbb{C}$. If F is reducible, R_F is the product of the polynomials of the strongly connected components of G_F.*

Proof. Al Ahmadieh [1, Lemma 5.1] proves, over any field and for every order, that a square matrix A is irreducible exactly when $\det(\operatorname{diag} x + A)$ is an irreducible polynomial in x. Take the field $\mathbb{C}$ and $A = -Q_F$. Then $R_F(w) = \det(\operatorname{diag}(-w) + A)$, and the linear change of variables $x = -w$ preserves irreducibility. For a reducible face, $-Q_F$ is block triangular with the blocks of the components on the diagonal, and the determinant factors accordingly. □

The proof of Al Ahmadieh rests on a theorem of Hartfiel and Loewy [16, Theorem 1], that every entry of the adjugate of $\operatorname{diag} x + A$ is a nonzero polynomial when A is irreducible.

Lemma 3.2 (the Perron branch)**.** *Let F be irreducible, let Q' be another generator whose face F is irreducible, and let $c \in \mathbb{R}$. Suppose that $\lambda'_F(v) = \lambda_F(v) + c$ for all v in a nonempty open set $\mathcal{U} \subseteq \mathbb{R}^F$. Then $-Q'_F$ and $-Q_F + cI$ have the same principal minors, and $\lambda'_F = \lambda_F + c$ on all of $\mathbb{R}^F$. In particular, for $c = 0$, $-Q'_F$ and $-Q_F$ have the same principal minors.*

Proof. Let $k = |F|$ and number the states of F as $1, \dots, k$. Put $\widetilde{R}(w) = \det(-Q_F + cI - \operatorname{diag} w) = R_F(w - c\mathbf{1})$, which is irreducible over $\mathbb{C}$ by Lemma 3.1. For $k = 1$ the hypothesis says $q'_1 = q_1 + c$, which is the claim. Let $k \geq 2$, and put $\psi(v) = v + \lambda'_F(v)\mathbf{1}$. For $v \in \mathcal{U}$ the number $\lambda'_F(v) = \lambda_F(v) + c$ is an eigenvalue of $-Q'_F - \operatorname{diag} v$ and of $-Q_F + cI - \operatorname{diag} v$, so $R'_F(\psi(v)) = 0 = \widetilde{R}(\psi(v))$.

Both polynomials are affine in w_1. Write $\widetilde{R} = p_1 w_1 + p_0$ and $R'_F = p'_1 w_1 + p'_0$ with p_1, p_0, p'_1, p'_0 polynomials in $w' = (w_2, \dots, w_k)$. The coefficient of $\prod_{i \geq 2}(-w_i)$ in p_1 is -1, so $p_1 \neq 0$. The polynomial $\mathcal{D} = p_1 R'_F - p'_1 \widetilde{R} = p_1 p'_0 - p'_1 p_0$ does not involve w_1 and vanishes on $\psi(\mathcal{U})$. We claim that the projection of $\psi(\mathcal{U})$ to w' contains a nonempty open set. Fix $v^0 \in \mathcal{U}$ and consider $y \mapsto (\psi(v^0 + (0, y)))_{i \geq 2}$ for small $y \in \mathbb{R}^{k-1}$. By Proposition 2.3(b), λ'_F is real-analytic with $\partial \lambda'_F / \partial v_i = -\beta_i$, where $\beta = \alpha'^*_F(v) > 0$ and $\sum_i \beta_i = 1$. The derivative of the map at $y = 0$ is $I - \mathbf{1}\beta'^{\top}$ with $\beta' = (\beta_i)_{i \geq 2}$, of determinant $\beta_1 > 0$, and the inverse function theorem proves the claim. So $\mathcal{D} = 0$ and $p_1 R'_F = p'_1 \widetilde{R}$. The irreducible $\widetilde{R}$ divides $p_1 R'_F$, and it involves w_1, so it does not divide $p_1 \neq 0$, which is free of w_1. Hence it divides R'_F. Both have total degree k and the coefficient 1 at $\prod_i(-w_i)$, so $R'_F = \widetilde{R}$. Finally $\lambda'_F(v)$ is the smallest real root of $t \mapsto R'_F(v + t\mathbf{1}) = \widetilde{R}(v + t\mathbf{1})$, which is $\lambda_F(v) + c$. □

Delmas, Dronnier and Zitt [10, Theorem 2.1] prove a multiplicative analogue for $\eta \mapsto \varrho(K \operatorname{diag} \eta)$ on $\{0,1\}^n$, by an induction over the cube. Cohen [9] proved that $v \mapsto \varrho(A + \operatorname{diag} v)$ is convex for a Metzler matrix A, and Friedland [14, Theorem 4.1] showed that for irreducible A it is affine only along $\mathbf{1}$. Lemma 3.2 is the inverse statement, that the map on an open set determines the principal minors of A. We did not find it in these papers.

3.2. The functional determines exactly the principal minors

Theorem 3.3 (the functional and the minors)**.** *For two irreducible generators Q and Q' on $[d]$ the following are equivalent.*

(i) $J_V = J'_V$ *on* Δ_V.

(ii) $\lambda_V = \lambda'_V$ *on a nonempty open set of fields.*

(iii) $-Q$ *and* $-Q'$ *have the same principal minors.*

If they hold, then $\lambda_F = \lambda'_F$ *on* $\mathbb{R}^F$ *and* $J_F = J'_F$ *on* Δ_F *for every face* F*, and* F *is irreducible for* Q *exactly when it is irreducible for* Q'.

Proof. (i) implies (ii). By Proposition 2.3(b), $\lambda_V(v) = \min_{\Delta_V}(J_V - v \cdot \alpha) = \lambda'_V(v)$ for every v.

(ii) implies (iii). This is Lemma 3.2 with $F = V$ and $c = 0$.

(iii) implies (i). Equal principal minors give $R'_F = R_F$ for every face F, since the minors of $-Q_F$ are among those of $-Q$. By Lemma 3.1, F is irreducible for Q exactly when R_F is irreducible, which is the last claim. On an irreducible face $\lambda'_F(v)$ and $\lambda_F(v)$ are the smallest real roots of the same polynomial $t \mapsto R_F(v + t\mathbf{1})$, so they agree. For a reducible face, the factors of $R_F = R'_F$ given by Lemma 3.1 are irreducible, so by unique factorization the components of G_F have the same vertex sets for both generators, and $\lambda_F(v)$, the least $\lambda_C(v)$ over the components, agrees too. By Proposition 2.3(b), J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°, so $J'_F = J_F$ there for irreducible F. On the boundary we use that J_F is convex and lower semicontinuous, as a supremum of affine functions of α. For $\alpha \in \Delta_F$ and $\beta \in \Delta_F^\circ$ the points $\alpha_t = (1-t)\alpha + t\beta$ with $t \in (0, 1]$ are interior. Convexity gives $\limsup_{t \downarrow 0} J_F(\alpha_t) \le J_F(\alpha)$, and lower semicontinuity the reverse inequality. So $J_F(\alpha) = \lim_{t \downarrow 0} J_F(\alpha_t)$, likewise for J'_F, and $J'_F = J_F$ on Δ_F. With $F = V$ this is (i). For a reducible face, Proposition 2.3(a) gives $J_F = J_V = J'_V = J'_F$ on Δ_F. □

Remark 3.4 (the survival probability). A finer observation than J_V gives the same minors more directly. Let v be a field and let the start have full support. The weighted mass $\sum_n P_N(n) e^{Tv \cdot n/N}$ is the probability that the chain survives when it is killed at each visit to i with probability $1 - e^{Tv_i/N}$, for $v \le 0$. As $N \to \infty$ at fixed T it tends to $\mu^\top e^{T(Q + \operatorname{diag} v)}\mathbf{1}$. For any generator Q' the limit $-T^{-1} \log \mu^\top e^{T(Q' + \operatorname{diag} v)}\mathbf{1}$ as $T \to \infty$ is $\lambda'_V(v) = -\varrho(Q' + \operatorname{diag} v)$, the least exit rate of the components of Q', because the start and the vector $\mathbf{1}$ are positive. So the survival probabilities on all fields give λ'_V on $\mathbb{R}^d$. Suppose that they agree with those of the irreducible Q. Then Q' is irreducible. Otherwise $\lambda'_V = \lambda'_C$ for some component $C \ne V$ on an open set of fields, and there λ'_V does not depend on v_j for $j \notin C$, while $\partial \lambda_V / \partial v_j = -\alpha_V^*(v)_j < 0$ by Proposition 2.3(b). Now Lemma 3.2 applies to Q and the irreducible Q' and gives every principal minor at once, with no descent through the faces.

3.3. The generators with given minors

Without reversibility the exit rates and the edge products $q_{ij}q_{ji}$ do not determine Q, since Q^* has the same exit rates, the same products and the same stationary law. The next theorem describes all generators with the principal minors of $-Q$.

Let C be a Z-matrix, that is $-C$ is Metzler, and let X be a cut of C with $Y = V \setminus X$. If C is irreducible, the blocks $C[X, Y]$ and $C[Y, X]$ are nonzero with nonpositive entries and

rank one, so $C[X,Y] = -ab^\top$ and $C[Y,X] = -cu^\top$ with $a, b, c, u \geq 0$. The cut-transpose of C across X is

$$C = \begin{pmatrix} C_{XX} & -ab^\top \\ -cu^\top & C_{YY} \end{pmatrix} \mapsto C' = \begin{pmatrix} C_{XX} & -ac^\top \\ -bu^\top & C_{YY}^\top \end{pmatrix}. \tag{4}$$

Another choice of the factors changes C' by a positive diagonal similarity $\operatorname{diag}(\alpha I_X, I_Y)$.

Theorem 3.5 (the generator fiber)**.** *Let Q be an irreducible generator.*

(a) Every generator Q' with the principal minors of $-Q$ is irreducible and has the spectrum of Q. The set of such generators is finite, and it contains Q and its time reversal Q^*.

(b) If Q has no cut, that is, for every partition $V = X \cup Y$ with $|X|, |Y| \geq 2$ one of the blocks $Q[X,Y]$, $Q[Y,X]$ has rank at least 2, then the set in (a) *is $\{Q, Q^*\}$. It is a single point exactly when Q is reversible.*

(c) In general the set in (a) *is the set of generator points of the cut-transpose orbit of $-Q$, in the following sense. Let C be an irreducible Z-matrix with the principal minors of $-Q$ and let X be a cut of C. Then the cut-transpose C' is again an irreducible Z-matrix with the principal minors of $-Q$, and exactly one positive diagonal similarity, up to a scalar, turns $-C'$ into a generator. Fewer than $2d$ cut-transposes, each across a cut of the current matrix, and one optional transposition reach every point of the set. If Q is reversible, the set is $\{Q\}$.*

(d) If $d \leq 3$, the set in (a) *is $\{Q, Q^*\}$ with no condition.*

Clause (c) needs the principal minors of $-Q$. For a general irreducible Z-matrix no diagonal similarity need produce a generator. The matrix $3I - P$ with P irreducible and stochastic is nonsingular, while the negative of every generator has the eigenvalue 0, and similarity preserves the spectrum.

What is borrowed. The matrix theorems used are Loewy [20, Theorem 1] for order $n \geq 4$ without a cut, Hartfiel and Loewy [16, Theorem 3] for $n \leq 3$, combined in [7, Lemma 2.8], Al Ahmadieh [1, Theorem 4.1 and Lemma 4.5] for the converse, and Chatterjee, Ghosh, Gurjar and Raj [7, Theorem 1.1] for the whole fiber. So (b), (d) and the reversible case of (c) follow from them together with the normalization (N) below. What we add is (N) and its consequences, the finiteness, the irreducibility of every member, and the blockwise description of Proposition 3.8.

Proof. We first record 2 facts used in (a) to (d).

(N) The generator point. Let C be an irreducible Z-matrix with the principal minors of $-Q$. The characteristic polynomial $\det(tI - C) = \sum_{s=0}^{d} (-1)^s t^{d-s} E_s(C)$, with E_s the sum of the principal minors of order s, is that of $-Q$. So $-C$ has the spectrum of Q, whose eigenvalue of largest real part is the simple eigenvalue 0. The theorem of Perron and Frobenius for the irreducible Metzler matrix $-C$ gives a vector $x > 0$, unique up to a factor, with $Cx = 0$. With $D = \operatorname{diag} x$ the matrix $-D^{-1}CD$ has off-diagonal entries $-c_{ij}x_j/x_i \geq 0$ and zero row sums, so it is an irreducible generator. If D' is real, diagonal and invertible and $-D'^{-1}CD'$ has zero row sums, then $C(D'\mathbf{1}) = 0$, so D' is a multiple of D and gives the same matrix.

We call $-D^{-1}CD$ the generator point of C. For $C = -Q$ it is Q, and for $C = -Q^\top$, whose null vector is π, it is Q^*.

(R) Real similarities. If $-Q' = E(-Q)E^{-1}$ or $-Q' = E(-Q)^\top E^{-1}$ for a complex invertible diagonal E, then E may be taken real. In the first case $q'_{ij} = e_i q_{ij}/e_j$, so $e_i/e_j = q'_{ij}/q_{ij}$ is real on every edge of the graph of Q. That graph is strongly connected, so every ratio e_i/e_1 is a product of such ratios and is real, and E/e_1 is a real diagonal similarity. The second case is the same with q_{ji} in place of q_{ij}.

(a) Let Q' be a generator with the principal minors of $-Q$, and suppose that $-Q'$ is reducible. Then there is a proper nonempty $A \subset V$ with $Q'[V \setminus A, A] = 0$, and $\det(-Q') = \det(-Q'_A)\det(-Q'_{V\setminus A})$. Both factors are proper principal minors, so they equal $\det(-Q_A)$ and $\det(-Q_{V\setminus A})$. By Lemma 2.2 every irreducible component of a proper face has a positive exit rate, so every eigenvalue of $-Q_A$ has positive real part and $\det(-Q_A) > 0$, and likewise $\det(-Q_{V\setminus A}) > 0$. This contradicts $\det(-Q') = \det(-Q) = 0$. So Q' is irreducible, and the same argument applies to every real matrix with the principal minors of $-Q$. This also follows from Lemma 3.1, since such a matrix has the polynomial R_V of $-Q$. The spectra agree by (N). The time reversal Q^* is a generator, since $Q^*\mathbf{1} = D_\pi^{-1}Q^\top\pi = 0$ and its off-diagonal entries are $\pi_j q_{ji}/\pi_i \geq 0$, and $-Q^* = D_\pi^{-1}(-Q)^\top D_\pi$ has the principal minors of $-Q$, since transposition and diagonal similarity preserve them. Finiteness follows from (c).

(b) For $d \leq 3$ the claim is (d). Let $d \geq 4$. Loewy's theorem [20, Theorem 1] applies to the irreducible matrix $-Q$ without a cut, with no condition on $-Q'$. It gives an invertible diagonal E with $-Q' = E(-Q)E^{-1}$ or $-Q'^\top = E(-Q)E^{-1}$, and by (R) we may take E real. In the first case (N) with $C = -Q$ gives $Q' = Q$, and in the second case $Q' = E^{-1}Q^\top E$ and (N) with $C = -Q^\top$ gives $Q' = Q^*$. Finally $Q^* = Q$ exactly when $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j, that is when Q is reversible.

(c) Let C and X be as in (c). The cut-transpose C' is a Z-matrix. It has the same principal minors as C. Indeed, for $U \subseteq V$ write $U_X = U \cap X$ and $U_Y = U \cap Y$. If U_X or U_Y is empty, both minors agree since a determinant is invariant under transposition. Otherwise replace C_{XX} by $C_{XX} + sI$ with s large, so that the block of U_X is invertible. The Schur complement gives

$$\det C_U = \det C_{U_X} \det\big(C_{U_Y} - \kappa\, c_{U_Y} b_{U_Y}^\top\big), \qquad \det C'_U = \det C_{U_X} \det\big(C_{U_Y}^\top - \kappa\, b_{U_Y} c_{U_Y}^\top\big),$$

with the same scalar $\kappa = u_{U_X}^\top C_{U_X}^{-1} a_{U_X}$. The two right sides are equal by transposition. Both sides are polynomials in s that agree for large s, so they agree at $s = 0$. By the argument of (a), C' is irreducible, and by (N) it has exactly one generator point.

Now let Q' be a generator in the fiber, irreducible by (a). The theorem of Chatterjee, Ghosh, Gurjar and Raj [7, Theorem 1.1] gives a sequence $-Q = C_0, C_1, \ldots, C_k$ with $k < 2d$, each C_{s+1} a cut-transpose of C_s across a cut of C_s [7, Definition 2.10], such that $-Q'$ is diagonally similar to C_k or to $C_k^\top$. Their cut-transpose uses a particular rank-one factorization, and another choice changes it by a real diagonal similarity $\mathrm{diag}(\alpha I_X, I_Y)$ [7, Remark 2.11]. A real invertible diagonal similarity of C by $\mathrm{diag}(E_X, E_Y)$ changes the cut-transpose by $\mathrm{diag}(E_X, E_Y^{-1})$, as a short computation with the blocks shows, and does not change the cuts. So, up to real invertible diagonal similarity, which (N) accepts, every C_s is obtained by our cut-transposes, and Q' is the generator point of C_k or of $C_k^\top$. Conversely every such generator point lies in the fiber. There are fewer than 2^d cuts at each step and fewer than $2d$ steps, so the fiber is finite, which completes (a).

If Q is reversible, $S = D_{\sqrt{\pi}}(-Q)D_{\sqrt{\pi}}^{-1}$ is an irreducible symmetric Z-matrix, and a symmetric matrix is its own cut-transpose. Indeed, if $S[X,Y] = -ab^\top$, then $S[Y,X] = -ba^\top$, so we may take $c = b$ and $u = a$. By the rule for diagonal similarities above, the cut-transpose of a matrix diagonally similar to S is again diagonally similar to S, and so is the transpose $E^{-1}SE$ of ESE^{-1}. Since $C_0 = -Q$ is diagonally similar to S, so is every C_s. Hence $-Q'$ is diagonally similar to $-Q$, and (N) with $C = -Q$ gives $Q' = Q$. See also Engel and Schneider [12, Theorem 3.5] and Al Ahmadieh [1, Lemma 6.1 and Theorem 6.2].

(d) For $d = 2$ a generator is fixed by q_1 and q_2, which are principal minors. Let $d = 3$ and let Q' be in the fiber, so Q' is irreducible by (a). The theorem of Hartfiel and Loewy [16, Theorem 3], as restated in [7, Lemma 2.8], says that for an irreducible matrix A of order 2 or 3 over a field, every matrix with the principal minors of A is diagonally similar to A or to $A^\top$ over that field. So $-Q' = E(-Q)E^{-1}$ or $-Q' = E(-Q)^\top E^{-1}$ with E diagonal and invertible, real by (R), and (N) gives $Q' = Q$ or $Q' = Q^*$, as in (b). □

A cut-transpose need not be Q or Q^*. Take $d = 4$, $X = \{1,2\}$ and

$$Q = \begin{pmatrix} -4.7 & 0.7 & 1 & 3 \\ 1.3 & -9.3 & 2 & 6 \\ 2 & 2 & -4.4 & 0.4 \\ 1 & 1 & 2 & -4 \end{pmatrix},$$

whose blocks $Q[X,Y] = (1,2)^\top(1,3)$ and $Q[Y,X] = (2,1)^\top(1,1)$ have rank one. To 4 decimals, the generator point of the cut-transpose of $-Q$ across X is

$$Q' = \begin{pmatrix} -4.7 & 0.7 & 2.3810 & 1.6190 \\ 1.3 & -9.3 & 4.7619 & 3.2381 \\ 0.84 & 0.84 & -4.4 & 2.72 \\ 1.8529 & 1.8529 & 0.2941 & -4 \end{pmatrix}.$$

Its entries are rational, for instance $q'_{13} = 50/21$, $q'_{41} = 63/34$ and $q'_{43} = 5/17$. It has the 15 principal minors of $-Q$, with $Q'_X = Q_X$ and $Q'_Y = Q^*_Y$, but $Q' \neq Q^*$, since $q^*_{12} = 351/530 \approx 0.6623$.

Remark 3.6 (three states by hand). For $d = 3$ the minors give the exit rates, the products $q_{ij}q_{ji}$, and the sum and product of the two oriented 3-cycle products $\omega = q_{12}q_{23}q_{31}$ and $\omega' = q_{13}q_{32}q_{21}$. So Q' has the cycle products of Q or of Q^*, hence differs from it by a potential, and (N) gives (d). Example 5.4 of Kim and Landim [18] is the case $\omega = 1$, $\omega' = 0$.

3.4. What the exact law adds

The law of the counts is finer than J_V, and for a generic start it separates the fiber. From the stationary start it cannot separate Q from Q^* at all.

Proposition 3.7 (the stationary start)**.** *Let Q be irreducible with stationary law π. With the start π, the law of the composition is the same under Q and under Q^*, for every N and every T with $Tq_{\max} \le N$, and so is the weighted law $P_N(n)e^{Tv\cdot n/N}$ for every field v.*

Proof. The matrix $P = I+(T/N)Q$ has stationary law π, and $P^* = D_\pi^{-1}P^\top D_\pi = I+(T/N)Q^*$ is stochastic, since Q^* has the exit rates of Q. For a path $x_1 \cdots x_N$ we have $\pi_{x_1}\prod_t P(x_t, x_{t+1}) = \pi_{x_N}\prod_t P^*(x_{t+1}, x_t)$. Reversal is a bijection of the paths that keeps the composition. Summing over the paths with composition n gives the claim, since the factor $e^{Tv\cdot n/N}$ depends only on n. □

For a face F that is irreducible for Q and a field v, the face maximum has the constant $K_F(v)$ of Definition 2.6. Let r and l be the right and left Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$. By Proposition 2.7,

$$K_F(v) = \frac{a_F(v)}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_{\alpha_F^*(v)}}}, \qquad a_F(v) = (\mu_F \cdot r)(l \cdot \mathbf{1}), \tag{5}$$

and we call $a_F(v)$ the amplitude.

Proposition 3.8 (the amplitudes)**.** *Let Q be irreducible, let $Q' \neq Q$ be a generator in the fiber of Theorem 3.5, and let F be a face that is irreducible for Q.*

(a) F is irreducible for Q', the two generators have the same λ_F, α_F^*, J_F and $\det \Sigma_{\alpha_F^*(v)}$, and $K'_F(v)/K_F(v) = a'_F(v)/a_F(v)$.

(b) There is a finite union $\mathcal{Z}$ of proper linear subspaces of $\mathbb{R}^V$ with the following property. For every start μ of full support outside $\mathcal{Z}$ and every $Q' \neq Q$ in the fiber, $K'_V(v) \neq K_V(v)$ for all v outside a closed nowhere dense set of fields. That set always contains $v = 0$, where every generator has $r = \mathbf{1}$, $l = \pi$ and $a_V(0) = 1$. If Q is not reversible, then $\pi \in \mathcal{Z}$.

(c) If Q' is the generator point of one cut-transpose of $-Q$ across X, then for every start $K'_F = K_F$ on every face $F \subseteq X$ and $K'_F = K^*_F$ on every face $F \subseteq Y$, where K^* belongs to Q^*.

Proof. (a) The first claims are Theorem 3.3. So $\alpha_F^*(v)$ and, by Proposition 2.7, $\det \Sigma$ at that point agree for Q and Q', and (5) gives the ratio.

(b) Take $F = V$ and put $g(v) = r(v)\,(l(v) \cdot \mathbf{1})$, which does not depend on the scale of r, so that $a_V(v) = \mu \cdot g(v)$. The map g is real-analytic on $\mathbb{R}^V$. Let W be the span of the vectors $g'(v) - g(v)$ over $v \in \mathbb{R}^V$. If $\mu \notin W^\perp$, the analytic function $a'_V - a_V = \mu \cdot (g' - g)$ is not identically zero, so its zero set is closed and nowhere dense. We show that $W \neq 0$. Suppose $g' = g$. Then $r'(v)$ is a multiple of $r(v)$, and both matrices have the eigenvalue $\lambda_V(v)$, so $(Q' - Q)r(v) = 0$ for every v. Take $v = te_j$ with $t \to \infty$. Then $\lambda_V \leq q_j - t \to -\infty$, and row $i \neq j$ of the eigenvalue equation gives $(q_i - \lambda_V)r_i \leq q_{\max} \max r$, so $r(te_j)/\max r \to e_j$. Hence $(Q' - Q)e_j = 0$ for every j and $Q' = Q$. So for $Q' \neq Q$ the space $W^\perp$ is proper, and $\mathcal{Z}$ is the union of these spaces over the finitely many $Q' \neq Q$ in the fiber. At $v = 0$ every generator has $r = \mathbf{1}$ and $l = \pi$, so $a_V(0) = \mu(V) = 1$. For the time reversal, $-Q^* - \operatorname{diag} v = D_\pi^{-1}(-Q - \operatorname{diag} v)^\top D_\pi$, so $r^* = D_\pi^{-1} l$ and $l^* = D_\pi r$, with $l^* \cdot r^* = 1$. At $\mu = \pi$ this gives $a_V^*(v) = (\mathbf{1} \cdot l)(\pi \cdot r) = a_V(v)$ for all v, so $\pi \in \mathcal{Z}$ when $Q^* \neq Q$, in line with Proposition 3.7.

(c) Let $C = -Q$ and let C' be its cut-transpose across X, with the factors of (4), and put $s_0 = (b \cdot \mathbf{1})/(c \cdot \pi_Y)$. The rows X of $Q\mathbf{1} = 0$ give $-Q_X\mathbf{1} = (b \cdot \mathbf{1})a$, the columns Y

of $\pi^\top Q = 0$ give $-Q_Y^\top \pi_Y = (\pi_X \cdot a)b$, and summing the columns X of $\pi^\top Q = 0$ gives $(\pi_X \cdot a)(b \cdot \mathbf{1}) = (\pi_Y \cdot c)(u \cdot \mathbf{1})$. With these identities, $x = (\mathbf{1}_X, s_0 \pi_Y)$ satisfies $C'x = 0$. So $Q' = -D_x^{-1} C' D_x$, and its killed blocks are $Q'_X = Q_X$ and $Q'_Y = D_{\pi_Y}^{-1} Q_Y^\top D_{\pi_Y} = Q_Y^*$. The constant $K_F(v)$ of a face depends only on Q_F and μ_F, which gives the claim. □

Proposition 3.8 concerns the law, not the labels of Section 4 (see the open questions in Section 7).

4. Modal supports

This section proves Theorem B. We set up the modal-support map in Section 4.1, make the notion of equal boundaries precise in Section 4.2, and prove the descent in Section 4.3.

4.1. The killed chain and its modal support

Fix a field $v \in \mathbb{R}^d$ and a time scale $\tau > 0$, put $T = \tau L$, and weight each composition by $e^{Tv \cdot n/N}$. Along a path the weight is $\prod_t e^{T v_{X_t}/N}$. If $v \le 0$, this is the probability that the chain survives when it is killed at each visit to i with probability $1 - e^{Tv_i/N}$. So $P_N(n) e^{Tv \cdot n/N}$ is the probability of surviving with composition n, and its maximizers are the modes of the composition of the surviving chain, since a normalization does not move a mode. On a face the extra killing shifts the diagonal of the killed generator, $Q_F + \operatorname{diag} v_F$ in place of Q_F, and $\lambda_F(v)$ is the exit rate of F for the killed chain.

The shift $v \mapsto v + c\mathbf{1}$ multiplies the weighted law by the constant e^{Tc}, and since $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$ it moves every score $\Phi_{\tau,v}(F)$ by the same $-\tau c$. So it changes no winner, and every field may be taken negative. Throughout this section the field and the time scale are calibrated, that is, v and τ are known in the units of Q.

Proposition 4.1 (the controlled score law)**.** *Fix Q, a start μ with full support, a field v and $\tau > 0$, and let $T = \tau L$. For every admissible face F,*

$$\log M_F(T, v) \le -\Phi_{\tau,v}(F)\, L + O(\log L),$$

with equality up to $O(\log L)$ if F is irreducible, and for a singleton $M_{\{i\}}(T, v)$ is given exactly by (2). A reducible admissible face F has $\Phi_{\tau,v}(F) \ge \Phi_{\tau,v}(C) + 1$ for one of its strongly connected components C. Hence if F is the unique minimizer of $\Phi_{\tau,v}$ over the admissible faces, then F is irreducible and the support of every maximizer of the weighted law is F for all large N.

Proof. For large N we have $T_0 \le \tau L \le N^{1/3}$, so Theorem 2.5(a) gives the upper bound. An irreducible face satisfies (H) (Lemma 2.1(a)), and Theorem 2.8 gives the equality. Let F be reducible and admissible, with $k = |F|$. Then $\lambda_F(v)$ is the least $\lambda_C(v)$ over the components C of F, and a component C where it is attained has $|C| \le k - 1$. It is irreducible, hence admissible, and $\Phi_{\tau,v}(F) = \Phi_{\tau,v}(C) + k - |C| \ge \Phi_{\tau,v}(C) + 1$. So a unique minimizer F of $\Phi_{\tau,v}$ is irreducible. Let $\varepsilon > 0$ be the least gap $\Phi_{\tau,v}(G) - \Phi_{\tau,v}(F)$ over the other admissible faces G. Their weighted maxima are at most $N^{-\Phi_{\tau,v}(F) - \varepsilon + o(1)}$, that of F is at least $N^{-\Phi_{\tau,v}(F) - o(1)}$, and a face that is not admissible carries no mass. There are finitely many faces, so for large N every maximizer has support F. □

The upper bound holds up to the boundary of the face, and this is what lets a fixed field act on all compositions at once.

4.2. Strict regions and controlled boundaries

Let $\Omega = (0,\infty) \times \mathbb{R}^d$ be the space of pairs $\omega = (\tau, v)$, and fix a start of full support. For a generator Q let $\mathcal{S}_Q$ be the set of $\omega \in \Omega$ at which $\Phi_{\tau,v}$ has a unique minimizer over the admissible faces, and let $W_Q(\omega)$ be that minimizer. We call $\mathcal{S}_Q$ the strict set and W_Q the modal-support map.

Lemma 4.2 (strict regions)**.** *Let Q be a generator and μ a start of full support.*

(a) The strict set $\mathcal{S}_Q$ is open and dense in Ω, its complement is closed and nowhere dense, and W_Q is locally constant on $\mathcal{S}_Q$.

(b) At every $\omega \in \mathcal{S}_Q$ the face $W_Q(\omega)$ is irreducible, and it is the support of every maximizer of the weighted law for all $N \geq N_0(\omega)$.

(c) Both $\mathcal{S}_Q$ and W_Q are invariant under $(\tau, v) \mapsto (\tau, v + c\mathbf{1})$ for every real c.

Proof. (b) is Proposition 4.1 and (c) is the shift rule. For (a), each score $\Phi_{\tau,v}(F)$ is continuous in (τ, v), since a Perron root is continuous in the entries, and there are finitely many faces. So a strict inequality between the least score and all others persists near a point of $\mathcal{S}_Q$, which shows that $\mathcal{S}_Q$ is open and W_Q locally constant. By Proposition 4.1 a reducible admissible face never attains the least score. So the complement of $\mathcal{S}_Q$ lies in the union, over pairs of distinct irreducible faces $F \neq H$, of the closed sets Z_{FH} where $\Phi_{\tau,v}(F) = \Phi_{\tau,v}(H)$. Each Z_{FH} has empty interior. If $|F| \neq |H|$, the difference $|F| - |H| + \tau(\lambda_F(v) - \lambda_H(v))$ vanishes for at most one τ at fixed v. If $|F| = |H|$, pick $j \in F \setminus H$. Then $\lambda_H(v)$ does not depend on v_j, while $\partial\lambda_F/\partial v_j = -\alpha_F^*(v)_j < 0$ by Proposition 2.3(b), so the difference vanishes for at most one v_j when the other coordinates are fixed. A finite union of closed sets with empty interior has empty interior, which gives (a). □

Definition 4.3 (controlled boundaries)**.** Fix a start μ of full support. Two generators Q and Q' have the same controlled boundaries if $W_Q = W_{Q'}$ on $\mathcal{S}_Q \cap \mathcal{S}_{Q'}$.

By Lemma 4.2(a) the set $\mathcal{S}_Q \cap \mathcal{S}_{Q'}$ is open and dense, and its complement is closed and nowhere dense. So Q and Q' have the same controlled boundaries exactly when their modal-support maps agree outside a closed nowhere dense subset of Ω. Since W_Q and $W_{Q'}$ are locally constant there, the set where they differ is open. So Q and Q' fail to have the same controlled boundaries exactly when there is a nonempty open set of ω on which both maps have a strict winner and the winners differ. By Lemma 4.2(c) it is the same to ask for agreement only on the open quadrant $(0,\infty) \times (-\infty, 0)^d$, where every field is a killing rate, since every point of Ω is moved into the quadrant by a shift. Ties are never observed and never used. The definition involves no threshold. Thresholds enter in the next lemmas, each with what it depends on.

4.3. Suppression and exposure

The next lemma uses the bound of Collatz and Wielandt. For a Metzler matrix B and a positive vector u, the Perron root of B is at most $\max_i (Bu)_i/u_i$ [2].

Lemma 4.4 (suppression)**.** *Let Q be a generator, F a nonempty face, $\mathcal{K} \subset \mathbb{R}^F$ compact and $\eta > 0$. There is $\bar{v}_0$, depending only on Q, F, $\mathcal{K}$ and η, such that for every field v with $v_F \in \mathcal{K}$ and $v_j \le -\bar{v}_0$ for all $j \notin F$, and every face $G \not\subseteq F$, we have $\lambda_G(v) \ge \lambda_{G\cap F}(v) - \eta$ if $G \cap F \ne \emptyset$, and $\lambda_G(v) \ge \bar{v}_0$ if $G \cap F = \emptyset$. Every larger value of $\bar{v}_0$ also serves.*

Proof. Put $P = G \cap F$, $S = G \setminus F$ and $B = Q_G + \operatorname{diag} v_G$, whose Perron root is $-\lambda_G(v)$. If $P = \emptyset$, every row of B has off-diagonal sum at most q_s and diagonal $-q_s + v_s \le -q_s - \bar{v}_0$, so its row sums are at most $-\bar{v}_0$, and the bound of Collatz and Wielandt with $u = \mathbf{1}$ gives $\lambda_G(v) \ge \bar{v}_0$.

Let $P \ne \emptyset$ and $\varrho_P(v) = -\lambda_P(v)$, and let $\mathbf{J}_P$ have ones off the diagonal of P. For $\zeta > 0$ the Perron root of the irreducible Metzler matrix $Q_P + \operatorname{diag} v_P + \zeta \mathbf{J}_P$ is continuous in (ζ, v_P) and decreases to $\varrho_P(v)$ as $\zeta \downarrow 0$. By Dini's theorem some ζ makes it at most $\varrho_P(v) + \eta/2$ for all $v_F \in \mathcal{K}$. Let x be its Perron vector with $\max x = 1$, and $x_* > 0$ the least entry of x over $\mathcal{K}$. Test B with $u = x$ on P and $u = \epsilon$ on S, where $\epsilon = \eta x_*/(2q_{\max})$. For $p \in P$, since $\mathbf{J}_P x \ge 0$,

$$(Bu)_p \le \big((Q_P + \operatorname{diag} v_P)x\big)_p + \epsilon q_{\max} \le (\varrho_P + \tfrac{\eta}{2})x_p + \epsilon q_{\max} \le (\varrho_P + \eta)x_p.$$

For $s \in S$,

$$(Bu)_s \le \sum_{p\in P} q_{sp}x_p + \epsilon\Big(\sum_{s'\in S\setminus\{s\}} q_{ss'} - q_s + v_s\Big) \le q_{\max} - \epsilon\bar{v}_0 \le (\varrho_P + \eta)\epsilon$$

once $\bar{v}_0 \ge q_{\max}/\epsilon + \max_{\mathcal{K}} |\varrho_P| + \eta$. There are finitely many P and S, so one $\bar{v}_0$ serves for all of them, and so does every larger value, as in the case $P = \emptyset$. The Perron root of B is then at most $\varrho_P(v) + \eta$, that is $\lambda_G(v) \ge \lambda_P(v) - \eta$. □

For an irreducible face F with $k = |F| \ge 2$ and an irreducible face $H \subsetneq F$, singletons included, put

$$\rho_H(v) = \frac{k - |H|}{\lambda_H(v) - \lambda_F(v)}, \qquad v \in \mathbb{R}^F.$$

The denominator is positive by Lemma 2.2, and ρ_H is real-analytic, since the Perron roots of irreducible Metzler matrices are simple. Also $\Phi_{\tau,v}(F) < \Phi_{\tau,v}(H)$ exactly when $\tau > \rho_H(v)$. So as τ decreases, the first proper subface to tie with F is a maximizer of $\rho_H(v)$.

Lemma 4.5 (exposure by descent)**.** *Let Q be irreducible with a start of full support, and let F be an irreducible face with $|F| \ge 2$. There is a closed nowhere dense set $\mathcal{E} \subset \mathbb{R}^F$ such that every $v^0 \in \mathbb{R}^F \setminus \mathcal{E}$ has a unique partner G, the irreducible face $G \subsetneq F$ that maximizes $\rho_H(v^0)$. Put $\tau_1 = \rho_G$. There are an open neighbourhood $\mathcal{U}$ of v^0 with compact closure and numbers $\delta > 0$ and $\bar{v}_0$, depending on Q, F and v^0, such that on the open set*

$$O = \big\{(\tau, v) \in \Omega :\ v_F \in \mathcal{U},\ v_j < -\bar{v}_0 \text{ for } j \notin F,\ |\tau - \tau_1(v_F)| < \delta\big\}$$

the following hold.

(i) The points of O with $\tau > \tau_1(v_F)$ lie in $\mathcal{S}_Q$ with $W_Q = F$, and those with $\tau < \tau_1(v_F)$ lie in $\mathcal{S}_Q$ with $W_Q = G$.

(ii) On the hypersurface $\tau = \tau_1(v_F)$ we have $\lambda_F(v) = \lambda_G(v) - (|F| - |G|)/\tau_1(v_F)$.

For $F = V$ the condition on v_j is empty.

So every irreducible face shares a boundary of strict regions with a proper subface, on an open set of (τ, v), and the position of that boundary is a relation between the two exit rates. Proposition 4.6 turns this into identifiability.

Proof. Let $k = |F|$.

Generic partners. Let $H_1 \neq H_2$ be irreducible proper subfaces of F with some $j \in H_1 \setminus H_2$, and take $v = te_j$ with $t \to \infty$. Then λ_{H_2} stays fixed, $\lambda_F \leq \lambda_{H_1} \leq q_j - t$ since a Perron root is at least every diagonal entry, and $\lambda_F \geq -t$ by the bound of Collatz and Wielandt with $u = \mathbf{1}$. Hence $\rho_{H_2} \to 0$, while $\rho_{H_1} \geq (k - |H_1|)/q_j > 0$, as $q_j > 0$ on the irreducible face F. So $\rho_{H_1} - \rho_{H_2}$ is a nonzero real-analytic function, with a closed nowhere dense zero set. The set $\mathcal{E}$ where the maximum of ρ_H over H is attained twice is closed and lies in the finite union of these zero sets.

The boundary. Let $v^0 \notin \mathcal{E}$ and let G be the unique maximizer of $\rho_H(v^0)$. Choose $\delta > 0$ with $3\delta < \tau_1(v^0)$ and $\rho_H(v^0) < \tau_1(v^0) - 3\delta$ for every irreducible $H \subsetneq F$ other than G. By continuity these hold with 2δ in place of 3δ on an open set around v^0, and we take for $\mathcal{U}$ a ball around v^0 whose closure lies in that open set. Let $\tau_\pm$ be the largest and the least value of τ on the closure of O. As $\tau_1 > 2\delta$ on the compact closure of $\mathcal{U}$, we get $\tau_- > \delta$. On the closure of O we have $\tau - \rho_H \geq \delta$ and $\lambda_H - \lambda_F$ is bounded below by a positive constant, so $\Phi(H) - \Phi(F) = (\lambda_H - \lambda_F)(\tau - \rho_H) \geq \kappa$ for some $\kappa > 0$. Also $\Phi(F) - \Phi(G) = (\lambda_G - \lambda_F)(\tau_1(v_F) - \tau)$ is negative above the hypersurface and positive below it, and a reducible face loses to one of its components (Proposition 4.1). So among the faces inside F the least score is attained only by F above and only by G below.

It remains to remove the faces $G' \not\subseteq F$. Apply Lemma 4.4 with the closure of $\mathcal{U}$ as $\mathcal{K}$ and $\eta = 1/(2\tau_+)$, which gives a threshold $\bar{v}_1$. If $P = G' \cap F$ is nonempty and C is a component of P with $\lambda_C(v) = \lambda_P(v)$, then $\Phi(G') \geq |P| + \tau\lambda_P(v) - \frac{1}{2} \geq \Phi(C) + \frac{1}{2}$, and C is an irreducible face inside F. Let $\bar{v}_0 \geq \bar{v}_1$ be so large that $\tau_-\bar{v}_0$ exceeds by $\frac{1}{2}$ the largest score of F on the closure of O, a bound that depends only on Q, F and $\mathcal{U}$. As $\bar{v}_0$ also serves in Lemma 4.4, an empty P gives $\Phi(G') \geq \tau_-\bar{v}_0$. Then every $G' \not\subseteq F$ scores at least $\frac{1}{2}$ above the least score, which proves (i). The tie $\Phi(F) = \Phi(G)$ on the hypersurface is (ii). □

Proposition 4.6 (identifiability)**.** *Let Q be irreducible, let the start have full support, and let Q' be a generator with the same controlled boundaries. Then Q' is irreducible and $-Q'$ has the principal minors of $-Q$. Equivalently, if Q and Q' are irreducible and $-Q$ and $-Q'$ have different principal minors, then there is a nonempty open set of (τ, v) on which both have a strict winner and the two winners differ.*

Proof. The two forms are equivalent by the remark after Definition 4.3. We prove the first.

Irreducibility. By Lemma 2.2, at a fixed field v the face V is the strict winner for Q once τ is large, and by continuity on an open set of Ω. That open set meets the dense set $\mathcal{S}_{Q'}$, and at a common point $W_{Q'} = V$. A reducible face is never a unique minimizer (Proposition 4.1), so Q' is irreducible.

Singletons. Fix $i \neq 1$, let $\bar{\tau} = 1/(2\max(q_{\max}, q'_{\max}))$, and let I be a compact interval in $(0, \bar{\tau}]$ and J a compact interval that contains $q_i - q_1$ and $q'_i - q'_1$ in its interior. Let $\bar{v}_0$ be the larger of the thresholds of Lemma 4.4 for Q and Q', with $F = \{1, i\}$, the compact set

$[-1,1]\times([-1,1]+J)$ of (v_1,v_i) and $\eta=1/(4\bar\tau)$, enlarged as in the proof of Lemma 4.5. Consider the open set of points (τ,v) with τ in the interior of I, $|v_1|<1$, v_i-v_1 in the interior of J and $v_j<-\bar v_0$ for $j\notin\{1,i\}$. The pair $\{1,i\}$, if it is irreducible, has an exit rate at least $\min(q_1-v_1,q_i-v_i)-\sqrt{q_{1i}q_{i1}}$, the smaller eigenvalue of a 2×2 matrix with real spectrum, and $\sqrt{q_{1i}q_{i1}}\le q_{\max}$. So it scores at least $\frac12$ above the better singleton, for both generators, and Lemma 4.4 removes the faces that leave $\{1,i\}$. The singleton $\{i\}$ beats $\{1\}$ exactly when $v_i-v_1>q_i-q_1$. So the strict winners of Q are $\{1\}$ and $\{i\}$ on the two sides of the hyperplane $v_i-v_1=q_i-q_1$, and those of Q' on the two sides of $v_i-v_1=q'_i-q'_1$. If the two hyperplanes differed, the open slab between them would carry different strict winners. So $q'_i-q'_1=q_i-q_1$. Put $c=q'_1-q_1$. Then $\lambda'_{\{i\}}=\lambda_{\{i\}}+c$ for every i.

The induction. We show by induction on $|F|$ that every irreducible face F of Q is irreducible for Q' and that $\lambda'_F=\lambda_F+c$ on $\mathbb{R}^F$. Singletons are the base. Let $|F|\ge 2$, choose $v^0\notin\mathcal{E}$ with partner G, and take O from Lemma 4.5. The set $O\cap\mathcal{S}_{Q'}$ is dense in O, and there $W_{Q'}=W_Q$. So F is the unique minimizer of the scores Φ' of Q' at points of a dense subset of the part of O above the hypersurface, and G at points of a dense subset of the part below. In particular F and G are irreducible for Q'. By continuity of the scores, $\Phi'(F)\le\Phi'(G)$ on the closure of the upper part and $\Phi'(F)\ge\Phi'(G)$ on the closure of the lower part, so $\Phi'(F)=\Phi'(G)$ on the hypersurface. With (ii) of Lemma 4.5 and the induction hypothesis for G,

$$\lambda'_F(v)=\lambda'_G(v)-\frac{|F|-|G|}{\tau_1(v_F)}=\lambda_G(v)+c-\frac{|F|-|G|}{\tau_1(v_F)}=\lambda_F(v)+c\qquad(v_F\in\mathcal{U}).$$

Lemma 3.2 extends this to all of $\mathbb{R}^F$. For $F=V$ both generators have $\lambda_V(0)=\lambda'_V(0)=0$, since $\mathbf{1}$ is a positive null vector of Q and of Q'. So $c=0$, and Lemma 3.2 with $F=V$ and $c=0$ gives equal principal minors. □

The induction is an algorithm. For each face choose a field at random, decrease τ from a large value, and read the first change of label and its position. The proof uses the strict regions only on an open set where V wins, on the open sets of the singleton step and on the sets O of Lemma 4.5. Their thresholds depend on Q and Q'. So no experiment is fixed in advance, unlike the reversible design of Section 5, and the statement is one of identifiability.

Theorem 4.7 (the modal supports and the minors)**.** *Let Q and Q' be irreducible generators and let the start have full support. They have the same controlled boundaries if and only if $-Q$ and $-Q'$ have the same principal minors.*

Proof. One direction is Proposition 4.6. For the other, suppose the minors agree. By Theorem 3.3, Q and Q' have the same irreducible faces and the same λ_F on each, so the scores $\Phi_{\tau,v}$ of the irreducible faces agree. With a start of full support every irreducible face is admissible, and every reducible admissible face scores above one of its components (Proposition 4.1). So for both generators a point lies in the strict set exactly when the least score over the irreducible faces is attained once, and the strict winner is that face. Hence $\mathcal{S}_Q=\mathcal{S}_{Q'}$ and $W_Q=W_{Q'}$. □

5. A reversible design and finite horizons

5.1. The design

For a reversible chain the modal supports determine all of Q, and an explicit finite experiment reads it. Let Q be reversible and irreducible with stationary law π, on a known connected graph (V, E) with $m = |E|$ edges, where $\{i, j\} \in E$ exactly when $q_{ij} > 0$. We call it the graph E. Put $w_{ij} = q_{ij}q_{ji}$ for the edge products and

$$S = D_{\sqrt{\pi}}(-Q)D_{\sqrt{\pi}}^{-1}, \qquad S_{ii} = q_i, \qquad S_{ij} = -\sqrt{w_{ij}} \ \ (\{i, j\} \in E),$$

the symmetrized matrix. We assume known constants $R, b > 0$ with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on every edge. The field and the time are calibrated.

We use 2 facts throughout. First, S is symmetric and similar to $-Q$, whose eigenvalues have nonnegative real parts, so S is positive semidefinite. Hence $0 \le q_k \le R$ for every k, $\sqrt{w_{kl}} \le R$ on every edge, and $|q_k - q_l| \le R$. Second, for a face F on which E induces a connected graph, Proposition 2.3(c) gives

$$\lambda_F(v) = \lambda_{\min}\big(S_F - \operatorname{diag} v_F\big). \tag{6}$$

Since Q has symmetric support, the admissible faces are exactly these connected faces (Lemma 2.1).

The design uses, for a retained pair (i, j), the field

$$v_i = \xi, \qquad v_j = 0, \qquad v_k = -\bar{v} \ \ (k \ne i, j),$$

which suppresses every other state, with the constants

$$\bar{\xi} = 2R, \qquad \tau_0 = \frac{1}{2R}, \qquad U = \frac{8R}{b^2}, \qquad \bar{v} = 4R + 4UR^2,$$

and the ranges $|\xi| \le 3R$ and $\frac{1}{4R} \le \tau \le U$. Every constant is fixed by R and b.

Lemma 5.1 (suppression in the reversible design)**.** *Under the design for the pair (i, j), every irreducible face that contains a suppressed state has score at least $\frac{3}{4}$ larger than the best face inside $\{i, j\}$, uniformly on the ranges of ξ and τ. Hence the strict winner is one of $\{i\}$, $\{j\}$ and $\{i, j\}$, and the last only if $\{i, j\}$ is an edge.*

Proof. Let F be connected and let $A = F \setminus \{i, j\}$ be nonempty.

Wholly suppressed faces. If $F \cap \{i, j\} = \emptyset$, then $S_F - \operatorname{diag} v_F = S_F + \bar{v}I \ge \bar{v}I$, so $\lambda_F(v) \ge \bar{v}$ by (6). The singleton $\{j\}$ has score $\tau q_j \le \tau R$. Hence $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(\{j\}) \ge \tau(\bar{v} - R) \ge (3R + 4UR^2)/(4R) \ge \frac{3}{4}$.

Mixed faces. Let $G_0 = F \cap \{i, j\}$ be nonempty. In the order G_0, A,

$$S_F - \operatorname{diag} v_F = \begin{pmatrix} B & Z \\ Z^\top & S_A + \bar{v}I \end{pmatrix}, \qquad B = S_{G_0} - \operatorname{diag} v_{G_0}, \quad Z = S[G_0, A].$$

Here $\|Z\|_2 \le R$ and $S_A \ge 0$. Let $e = \lambda_{\min}(B)$, so $|e| \le \|S_{G_0}\|_2 + |\xi| \le 4R$. The test vector $(x, 0)$, with x a unit eigenvector of B for e, gives $\lambda_F(v) \le e$. For a unit vector (x, y),

$$\left\langle \begin{pmatrix} x \\ y \end{pmatrix}, (S_F - \operatorname{diag} v_F) \begin{pmatrix} x \\ y \end{pmatrix} \right\rangle \ge e|x|^2 - 2R|x||y| + \bar{v}|y|^2 \ge \lambda_{\min} \begin{pmatrix} e & -R \\ -R & \bar{v} \end{pmatrix} \ge e - \frac{R^2}{\bar{v} - e},$$

by $\sqrt{u^2+R^2} \le u + R^2/(2u)$ with $u = (\bar{v} - e)/2 > 0$. Since $\bar{v} - e \ge \bar{v} - 4R = 4UR^2$, we get $e - 1/(4U) \le \lambda_F(v) \le e$. Now $e = \lambda_{G_*}(v)$ for an irreducible face $G_* \subseteq \{i,j\}$. If G_0 is connected, take $G_* = G_0$ and use (6). If $G_0 = \{i,j\}$ is not an edge, B is diagonal and G_* is the singleton with the smaller diagonal entry. In both cases $|G_*| \le |F| - 1$, so $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(G_*) \ge 1 - \tau/(4U) \ge \frac{3}{4}$ as $\tau \le U$.

So the unique minimizer of $\Phi_{\tau,v}$, if there is one, is an irreducible face inside $\{i,j\}$, and Proposition 4.1 identifies it with the strict winner. □

Theorem 5.2 (reversible reconstruction)**.** *Let Q be reversible and irreducible on the known connected graph E with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on edges, and let the start have full support.*

(a) (Exit differences.) For each $i \ne 1$, under the design for the pair $(i,1)$ at $\tau = \tau_0$, the strict winner changes from $\{1\}$ to $\{i\}$ at $\xi = x_i = q_i - q_1$, and $|x_i| \le R$. Put $x_1 = 0$.

(b) (Products.) For each edge $\{i,j\}$, under the design for (i,j) at $\xi = \bar{\xi}$, the strict winner changes from a singleton to the pair at

$$\tau = t_{ij} = \frac{\sqrt{s_{ij}^2 + 4w_{ij}} + s_{ij}}{2w_{ij}}, \qquad s_{ij} = \bar{\xi} - x_i + x_j,$$

with $1/R \le t_{ij} < U/2$, and $w_{ij} = s_{ij}/t_{ij} + t_{ij}^{-2}$.

(c) (Reconstruction.) Let B be the symmetric matrix with diagonal x_i, entries $-\sqrt{w_{ij}}$ on the edges and zero elsewhere, let $\lambda = \lambda_{\min}(B)$ and let ϕ be its positive unit eigenvector. Then

$$S = B - \lambda I, \qquad Q = -D_\phi^{-1} S D_\phi, \qquad \pi_i = \phi_i^2.$$

So the $m + d - 1$ limiting boundary coordinates $(x_i)_{i \ne 1}$ and $(t_{ij})_{\{i,j\} \in E}$ determine Q, and the design is fixed by R, b and the labels before any boundary is observed.

Proof. By Lemma 5.1 and Proposition 4.1, at each point of the ranges the strict winner is the unique minimizer of $\Phi_{\tau,v}$ among the irreducible faces inside the retained pair, when it is unique. Here $\tau_0 \in [\frac{1}{4R}, U]$, since $b \le R$.

(a) The candidates are $\{i\}$, $\{1\}$ and, if $\{i,1\}$ is an edge, the pair. By Weyl's inequality $\lambda_{\{i,1\}}(v) \ge \min(q_i - \xi, q_1) - \sqrt{w_{i1}}$, so at τ_0 the score of the pair exceeds the better singleton score by at least $1 - \tau_0 R = \frac{1}{2}$, and the pair never wins. The singleton $\{i\}$ has the smaller score exactly when $q_i - \xi < q_1$, that is when $\xi > x_i$. Since $|x_i| \le R$, both sides of x_i lie in $[-3R, 3R]$. Hence $\{1\}$ wins strictly on $[-3R, x_i)$, $\{i\}$ wins strictly on $(x_i, 3R]$, and the boundary is $\xi = x_i$.

(b) Put $p = q_i - \bar{\xi}$ and $s = s_{ij}$. Since $x_i - x_j = q_i - q_j$ by (a), $s = q_j - p$. Then $R \le s \le 3R$, and $\{i\}$ beats $\{j\}$ by $\tau s > 0$. By (6) the pair has $\lambda_{\{i,j\}}(v) = p - g$, the smaller eigenvalue of $\left(\begin{smallmatrix} p & -\sqrt{w} \\ -\sqrt{w} & q_j \end{smallmatrix}\right)$ with $w = w_{ij}$, where

$$g = \frac{\sqrt{s^2 + 4w} - s}{2} = \frac{2w}{\sqrt{s^2 + 4w} + s} \in (0, \sqrt{w}\,].$$

So $\Phi(\{i,j\}) - \Phi(\{i\}) = 1 - \tau g$. Hence $\{i\}$ wins strictly below $t_{ij} = 1/g$, the pair wins strictly above it, and $1/g$ is the displayed formula. Also $g \le \sqrt{w} \le R$ gives $t_{ij} \ge 1/R >$

$\frac{1}{4R}$. The formula increases in s and decreases in w, and $s \le 3R$, $w \ge b^2$, $b \le R$, so $t_{ij} \le (\sqrt{13}+3)R/(2b^2) < U/2$. At $\tau = t_{ij}$ the pair eigenvalue $p - 1/t_{ij}$ is a root of $(p-y)(q_j-y) = w$, which gives $t_{ij}^{-1}(s + t_{ij}^{-1}) = w$, the inversion formula.

(c) $B = S - q_1 I$, since S has diagonal q_k, entries $-\sqrt{w_{kl}}$ on edges and zero off the graph. The positive unit vector $\sqrt{\pi}$ satisfies $S\sqrt{\pi} = D_{\sqrt{\pi}}(-Q)\mathbf{1} = 0$. As $S \ge 0$, the eigenvalue 0 is $\lambda_{\min}(S)$, and it is simple with eigenvector $\sqrt{\pi}$ by the theorem of Perron and Frobenius for the irreducible nonnegative matrix $RI - S$. So $\lambda = -q_1$, $S = B - \lambda I$, $\phi = \sqrt{\pi}$ and $\pi_i = \phi_i^2$, and $Q = -D_\phi^{-1} S D_\phi$ is the definition of S. The coordinates used are $d-1$ values x_i and m values t_{ij}, and the fields and time scales of the design depend only on R, b and the labels. □

The reconstruction of a reversible generator from its exit rates and edge products is classical. Kim and Landim [18, Lemma 5.1 and Corollary 5.3] recover the same quantities from the values of J_V, and with the stationary law from $J_V^{-1}(0)$ (their Lemma 5.2) deduce that J_V determines a reversible generator. Here the coordinates are read off the boundaries of strict regions of the weighted composition law, with a design fixed in advance.

Theorem 5.3 (minimality)**.** *The reversible generators on the fixed connected graph E form a real-analytic manifold of dimension $m + d - 1$, with global coordinates the edge products (w_{ij}) and the exit differences $(q_i - q_1)$. Let W be a nonempty open subset of it, for example the set where $\|S\|_2 < R$ and $\sqrt{w_{ij}} > b$ when that set is nonempty. Let $k < m+d-1$, let $f : W \to \mathbb{R}^k$ be C^1 and let $\ell : W \to \Lambda$ be locally constant with Λ finite. Then the observation (f, ℓ) is not injective on any nonempty open subset of W. So no family of fewer than $m + d - 1$ locally C^1 scalar observations with locally constant finite labels is locally identifying.*

Proof. Let $\mathcal{Q}_E$ be the set of reversible generators with $q_{ij} > 0$ exactly on the edges in E, and $\kappa(Q) = ((w_{ij})_E, (q_i - q_1)_{i \ne 1})$. Given $w \in (0,\infty)^m$ and $x \in \mathbb{R}^{d-1}$ (with $x_1 = 0$), form B as in Theorem 5.2(c), let λ and ϕ be its least eigenvalue and positive unit eigenvector, which is simple since $-B$ plus a large multiple of I is irreducible and nonnegative, and put $\Psi(w,x) = -D_\phi^{-1}(B - \lambda I)D_\phi$. This is a reversible generator on E, with respect to ϕ^2, with products w_{ij} and exit rates $x_i - \lambda$, so $\kappa \circ \Psi = \mathrm{id}$, and Theorem 5.2(c), whose proof uses only reversibility and irreducibility, gives $\Psi \circ \kappa = \mathrm{id}$. The map Ψ is real-analytic by the implicit function theorem at a simple eigenvalue, and κ is polynomial. So κ is a global real-analytic chart of $\mathcal{Q}_E$, of dimension $m + d - 1$.

For the lower bound let $W' \subseteq W$ be nonempty and open. Shrinking it, we may assume that ℓ is constant on W' and that W' is a ball in the chart κ, so f is a C^1 map on a ball of $\mathbb{R}^p$, $p = m + d - 1$. Let ϱ_f be the largest rank of df on W', attained at y_0. The rank is lower semicontinuous, so it equals ϱ_f near y_0, and by the constant rank theorem f is constant on a C^1 submanifold through y_0 of dimension $p - \varrho_f \ge p - k \ge 1$. So (f, ℓ) is not injective on W'. □

The chart is the continuous-time version of that of Wolfer and Watanabe [25, Theorems 3, 5 and 6] for reversible kernels, and the lower bound is the classical rank argument. Regularity is needed, since interleaving the digits of the $m + d - 1$ coordinates gives one injective scalar.

Remark 5.4 (partial data). Suppose that the products are known on $J \subseteq E$ and the exit differences along the edges of a comparison graph (V, E'). In the chart κ the generators with these data have the given w_e on J and any $w_e > 0$ on $E \setminus J$. Their x is the given one plus a

vector that is constant on each component of (V, E') and zero on that of the state 1. So they are parametrized by $(0,\infty)^{m-|J|} \times \mathbb{R}^{\mathrm{comp}(E')-1}$, with $\mathrm{comp}(E')$ the number of components of (V, E'), and the data identify Q exactly when $J = E$ and (V, E') is connected. For instance, for $\ell > \sqrt{2}$ and $z = \ell - 2/\ell$ the generators

$$Q_\ell = \begin{pmatrix} -\ell & 2/\ell & z \\ \ell/2 & -\ell & \ell/2 \\ z & 2/\ell & -\ell \end{pmatrix}$$

satisfy detailed balance with $\nu = (1, 4/\ell^2, 1)$, have all exit rates equal to ℓ and $w_{12} = w_{23} = 1$, and have $w_{13} = z^2$. So the data of a spanning tree miss a parameter.

Theorem 5.5 (stability)**.** *In the setting of Theorem 5.2, let the observed dimensionless coordinates x_i/R and Rt_{ij} have sup-norm errors at most ε, and let $\widehat{Q}$ be the reconstruction from the observed values. There are ε_0 and C depending only on d and b/R such that for $\varepsilon \le \varepsilon_0$ the reconstruction $\widehat{Q}$ is a reversible generator on E and*

$$\|\widehat{Q} - Q\|_2 \le C\, R\, \varepsilon.$$

Among the constants are the bound $\phi_{\min} \ge \phi_* = (b/R)^{d-1}/\sqrt{d}$ for the ground vector $\phi = \sqrt{\pi}$ and the bound $\lambda_2(S) \ge g_* = 2b\phi_*^2/(d-1)$ for the gap of S.

Proof. The reconstruction commutes with the scaling $Q \mapsto Q/R$, so we take $R = 1$, $\bar{\xi} = 2$ and $b \le 1$. Then $\sigma_{ij} = t_{ij} \ge 1$ and $s_{ij} = 2 - x_i + x_j \in [1, 3]$, by Theorem 5.2. Hats denote reconstructed values, and $\widehat{x}_1 = x_1 = 0$.

Products. Let $\varepsilon \le b^2/20$. Then $\widehat{\sigma} \ge \frac{3}{4}$ and $|\widehat{\sigma}^{-1} - \sigma^{-1}| \le \frac{4}{3}\varepsilon$, and $w = s/\sigma + \sigma^{-2}$ gives $|\widehat{w} - w| \le (\frac{20}{3} + \frac{28}{9})\varepsilon < 10\varepsilon$. So $\widehat{w} \ge b^2/2$ and $|\sqrt{\widehat{w}} - \sqrt{w}| \le 10\varepsilon/b$, and the largest absolute row sum gives $\|\widehat{B} - B\|_2 \le \beta := 10d\varepsilon/b$. By Weyl's inequality $|\widehat{\lambda} - \lambda| \le \beta$, so $\|\widehat{S} - S\|_2 \le 2\beta$. The matrix $\widehat{B}$ has negative entries exactly on the edges in E, so the chart Ψ of the proof of Theorem 5.3 makes $\widehat{Q}$ a reversible generator with the graph E.

The constants. Row l of $S\phi = 0$ reads $q_l\phi_l = \sum_{k\sim l} \sqrt{w_{lk}}\phi_k$, so $\phi_l \ge b\phi_k$ on every edge. Some ϕ_{k_0} is at least $1/\sqrt{d}$, and every state is joined to k_0 by at most $d-1$ edges, so $\phi_{\min} \ge \phi_*$. For $y \perp \phi$ write $y = \phi \circ f$. Then $\langle y, Sy\rangle = \sum_{\{k,l\}\in E} \chi_{kl}(f_k - f_l)^2$ with $\chi_{kl} = \sqrt{w_{kl}}\phi_k\phi_l \ge b\phi_*^2$, while $\|y\|^2 = \sum_k \pi_k f_k^2$ and $\sum_k \pi_k f_k = 0$. In a spanning tree $\mathcal{T}$ of E the path from k to l has at most $d-1$ edges, so the inequality of Cauchy and Schwarz gives

$$\|y\|^2 = \tfrac{1}{2}\sum_{k,l} \pi_k\pi_l(f_k - f_l)^2 \le \tfrac{d-1}{2}\sum_{e\in\mathcal{T}} (\nabla_e f)^2 \le \frac{d-1}{2b\phi_*^2}\langle y, Sy\rangle,$$

and $\lambda_2(S) \ge g_*$.

The generator. Write $\widehat{\phi} = a\phi + z$ with $z \perp \phi$ and $a > 0$. Since $Sz = (S - \widehat{S})\widehat{\phi}$, we get $\|z\| \le 2\beta/g_*$, and since $1 - a \le 1 - a^2 = \|z\|^2$, also $\gamma = \|\widehat{\phi} - \phi\| \le \sqrt{2}\|z\| \le 2\sqrt{2}\beta/g_*$. This is the bound of Davis and Kahan in its simplest form. Let ε be so small that $\gamma \le \phi_*/2$. Then $\widehat{\phi}_{\min} \ge \phi_*/2$, $|\widehat{\phi}_l/\widehat{\phi}_k - \phi_l/\phi_k| \le 4\gamma/\phi_*^2$ and $\widehat{\phi}_l/\widehat{\phi}_k \le 2/\phi_*$, so on an edge

$$|\widehat{q}_{kl} - q_{kl}| = \big|\sqrt{\widehat{w}_{kl}}\,\widehat{\phi}_l/\widehat{\phi}_k - \sqrt{w_{kl}}\,\phi_l/\phi_k\big| \le \frac{20\varepsilon}{b\phi_*} + \frac{4\gamma}{\phi_*^2}.$$

The diagonal satisfies $|\widehat{q}_k - q_k| = |\widehat{S}_{kk} - S_{kk}| \le 2\beta$, and off the graph both matrices vanish. So every entry of $\widehat{Q} - Q$ is at most $C_1(d,b)\varepsilon$, and $\|\widehat{Q} - Q\|_2 \le dC_1\varepsilon$. The conditions on ε depend only on d and b. □

The perturbation bounds are standard. The point is that the constants come from R and b alone, with no separate assumption on the spectral gap.

5.2. Finite horizons

Theorem 5.2 reads the coordinates off the limiting boundaries. At a finite horizon the observed boundaries are shifted, and the shift has an explicit form, uniformly over a class of chains and starts. Throughout, the connected graph E on $[d]$ and the numbers $0 < b \le R$ and $0 < \mu_* \le 1/d$ are fixed. Let $\mathcal{C}$ be the set of reversible irreducible generators on E with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on every edge, and let $\mathcal{M}$ be the set of starts with $\min_i \mu_i \ge \mu_*$. For a pair (i,j) and $\xi \in \mathbb{R}$ let $v^{ij}(\xi)$ be the design field of Section 5.1, and let $\mathcal{V}$ be the compact set of these fields with $|\xi| \le 3R$.

The fixed-chain face maxima are Theorem 2.8. The finite horizon also needs constants that do not depend on the chain and the start in the class. This is the last result we import from [22].

Proposition 5.6 (uniform face maxima on the class, [22, Proposition 5.12])**.** *Let E induce a connected graph on the face F, with $k = |F| \ge 2$. There are N_0 and C, depending only on E, R, b and μ_*, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $v \in \mathcal{V}$ the following hold.*

(a) For all $N \ge 2$ and $T > 0$ with $TR \le N/2$, $M_F(T,v) \le C(T/N)^{k-1} e^{-T\lambda_F(v) + CT^2/N}$.

(b) For $N \ge N_0$ and $1 \le T \le N^{1/3}$, we have $C^{-1} \le K_F(v) \le C$ and

$$\left| \frac{M_F(T,v)}{K_F(v)\, N^{-(k-1)} T^{(k-1)/2} e^{-T\lambda_F(v)}} - 1 \right| \le C\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

For a singleton, $M_{\{i\}}(T,v)$ is given exactly by (2).

The proof in [22] is a compactness argument. On an edge $q_{ij} \ge b^2/R$, so the class, the starts and the fields form a compact set on which every connected face stays irreducible, and every constant in the fixed-chain proofs of Lemma 2.4 and Theorem 2.8 is continuous there.

Theorem 5.7 (finite-horizon recovery on the class)**.** *Run the design of Theorem 5.2 at horizon N, with $L = \log N$ and $T = \tau L$, and let $\mathcal{W}_N(\tau, v)$ be the set of supports of the maximizers of $n \mapsto P_N(n) e^{Tv\cdot n/N}$. For an ordered pair (i,j) with $\{i,j\} \in E$ put $v = v^{ij}(\bar{\xi})$,*

$$\tau_N^- = \sup\Big\{\tau \in [\tfrac{1}{4R}, U] : \mathcal{W}_N(\tau', v) = \{\{i\}\} \text{ for all } \tau' \in [\tfrac{1}{4R}, \tau]\Big\},$$

$$\tau_N^+ = \inf\Big\{\tau \in [\tfrac{1}{4R}, U] : \mathcal{W}_N(\tau', v) = \{\{i,j\}\} \text{ for all } \tau' \in [\tau, U]\Big\},$$

let $t_{ij,N}$ be any number in $[\tau_N^-, \tau_N^+]$, and let $K_{ij} = K_{\{i,j\}}(v)$. Put $\varepsilon_N = \log L/(2L)$. There are N_0 and C, depending only on E, R, b and μ_*, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $N \ge N_0$ the following hold.

(a) (Singleton switches.) Let $i \neq 1$. For every $\xi \in [-3R, 3R]$ other than

$$x_{i,N} = \frac{1}{\tau_0 L}\Big[\log\frac{\mu_1}{\mu_i} + (N-1)\log\frac{N - \tau_0 L q_1}{N - \tau_0 L q_i}\Big],$$

$\mathcal{W}_N(\tau_0, v^{i1}(\xi))$ is $\{\{1\}\}$ for $\xi < x_{i,N}$ and $\{\{i\}\}$ for $\xi > x_{i,N}$. Moreover $|x_{i,N} - x_i - 2R\log(\mu_1/\mu_i)/L| \le CL/N$, so $|x_{i,N} - x_i| \le 2R\log(1/\mu_*)/L + CL/N$.

(b) (Pair entries.) We have $\tau_N^+ - \tau_N^- \le C/L^2$ and

$$\Big|t_{ij,N} - t_{ij}(1-\varepsilon_N) - \frac{c_{ij}}{L}\Big| \le C\,\frac{\log L}{L^2}, \qquad c_{ij} = -t_{ij}\Big(\frac{1}{2}\log t_{ij} + \log\frac{K_{ij}}{\mu_i}\Big), \qquad |c_{ij}| \le C.$$

(c) (Reconstruction.) Let $\widehat{Q}_N$ be the reconstruction of Theorem 5.2(c) from the values $x_{i,N}$ and $t_{ij,N}$, and $\widetilde{Q}_N$ the one from $x_{i,N}$ and $t_{ij,N}/(1-\varepsilon_N)$. Both are reversible generators on E, and

$$\|\widehat{Q}_N - Q\|_2 \le C\,\frac{\log L}{L}, \qquad \|\widetilde{Q}_N - Q\|_2 \le \frac{C}{L}.$$

The threshold N_0 is large. The design runs up to $\tau = U = 8R/b^2$, and $I + (T/N)Q$ is stochastic on the whole range exactly when $N \ge U q_{\max}\log N$. So the least such N has $Nb^2/\log N = 8Rq_{\max}$ up to rounding, and it grows like $b^{-2}\log(1/b)$ when $q_{\max}$ stays of order R. Section 7 computes it for 2 families. The proof compares the exact singleton atoms with the uniform face maximum of the pair and solves the resulting crossing equation for T.

Proof. All constants below depend only on E, R, b and μ_*. On the ranges of the design $T \in [L/(4R), UL]$, so for $N \ge N_1$ we have $TR \le N/2$, $T \le N^{1/3}$ and $1/T + T^2/N \le 5R/L$, and Proposition 5.6 applies.

Suppression. Fix a pair and a point of the ranges, and let m^* be the least score among the irreducible faces inside the pair. The proof of Lemma 5.1 uses only R, b and $\bar{v}$, so each connected face G' outside the pair has $\Phi_{\tau,v}(G') \ge m^* + \frac{3}{4}$, and Proposition 5.6(a) or the singleton atom gives $M_{G'} \le CT^{|G'|-1}N^{-\Phi_{\tau,v}(G')}$. The best face inside the pair has face maximum at least $C^{-1}N^{-m^*}$, by the singleton atom with $\mu_i \ge \mu_*$ or by Proposition 5.6(b). Faces that are not connected carry no mass. So for $N \ge N_0$ every maximizer has its support inside the pair.

(a) At τ_0 the pair $\{i,1\}$, if it is an edge, has score at least the better singleton score plus $\frac{1}{2}$, by the proof of Theorem 5.2(a), so as above its face maximum is smaller than $\max(M_{\{1\}}, M_{\{i\}})$ by a factor $CLN^{-1/2}$. The two singleton atoms are exact, and

$$\log M_{\{i\}} - \log M_{\{1\}} = \log\frac{\mu_i}{\mu_1} + (N-1)\log\frac{N - \tau_0 L q_i}{N - \tau_0 L q_1} + \tau_0 L\,\xi,$$

which is affine and increasing in ξ with zero $x_{i,N}$. This gives the claim on $\mathcal{W}_N$. Since $|\log(1-y)+y| \le y^2$ for $0 \le y \le \frac{1}{2}$, the middle term is $-\tau_0 L(q_i - q_1) + O(L/N) + O(R^2L^2/N)$, and dividing by $\tau_0 L = L/(2R)$ gives the expansion. For large N it gives $x_{i,N} \in (-3R, 3R)$, since $|x_i| \le R$.

(b) Let $\{i,j\}$ be an edge and $v = v^{ij}(\bar{\xi})$. The exact atoms give $\log M_{\{j\}} - \log M_{\{i\}} \le \log(1/\mu_*) - TR + 1$ with $TR \ge L/4$, so $\{j\}$ loses to $\{i\}$. By the proof of Theorem 5.2(b),

$\lambda_{\{i\}}(v) - \lambda_{\{i,j\}}(v) = 1/t_{ij}$, and the exact atom gives $\log M_{\{i\}} = \log \mu_i - T(q_i - \bar{\xi}) + O(1/L)$. So Proposition 5.6(b) with $k = 2$ gives, on the whole range,

$$\Lambda_N(T) = \log M_{\{i,j\}} - \log M_{\{i\}}, \qquad \big|\Lambda_N(T) + L - \Xi(T) - \kappa\big| \le \frac{C_5}{L},$$

with $\Xi(T) = T/t_{ij} + \frac{1}{2}\log T$ and $\kappa = \log(K_{ij}/\mu_i)$. Here $|\kappa| \le C$, since K_{ij} is bounded above and below and $\mu_i \ge \mu_*$. The function Ξ increases with $\Xi' \ge 1/t_{ij} \ge 2/U$. Let T° solve $\Xi(T^\circ) = L - \kappa$. Then $\Lambda_N(T) < 0$ for $T \le T^\circ - C_5 U/L$ and $\Lambda_N(T) > 0$ for $T \ge T^\circ + C_5 U/L$. As $t_{ij} \in [1/R, U/2)$, the window $|T - T^\circ| < C_5 U/L$ lies inside the range for large N. With the suppression step, every maximizer has support $\{i\}$ below the window and $\{i,j\}$ above it. Hence $T^\circ/L - C_5 U/L^2 \le \tau_N^- \le \tau_N^+ \le T^\circ/L + C_5 U/L^2$, which is the first claim, and $t_{ij,N} = T^\circ/L + O(1/L^2)$. To solve for T° put $\tau^\circ = T^\circ/L$. Then $\tau^\circ = t_{ij}\big(1 - (\frac{1}{2}\log T^\circ + \kappa)/L\big)$, so $\tau^\circ/t_{ij} = 1 + O(\log L/L)$ and $\log T^\circ = \log L + \log t_{ij} + O(\log L/L)$. Putting this back gives $\tau^\circ = t_{ij}(1 - \varepsilon_N) + c_{ij}/L + O(\log L/L^2)$, uniformly, since $t_{ij} \le U/2$ and $|\kappa| \le C$.

(c) By (a) and (b) the dimensionless coordinates $x_{i,N}/R$ and $Rt_{ij,N}$ differ from x_i/R and Rt_{ij} by at most $C \log L/L$, and after the correction by at most C/L, since $t_{ij,N}/(1-\varepsilon_N) - t_{ij} = (c_{ij}/L + O(\log L/L^2))/(1-\varepsilon_N)$. Theorem 5.5 gives its admissible error size ε_0 and its constant as functions of d and b/R alone. So for $N \ge N_0$ these errors are below ε_0, and that theorem gives both bounds and the generator property. □

Remark 5.8 (a start that defeats every horizon). The bound $\min_i \mu_i \ge \mu_*$ cannot be dropped, since the start enters through the singleton switches as $2R\log(\mu_1/\mu_i)/L$. Let $d = 2$, $q_{12} = q_{21} = 1$, $\tau = \frac{1}{4}$ and $v = (0,1)$. The scores are $\Phi(\{2\}) = 0$, $\Phi(\{1\}) = \frac{1}{4}$ and $\Phi(\{1,2\}) = 1 + \frac{1}{4}\cdot\frac{1-\sqrt{5}}{2} > \frac{3}{4}$, so for a fixed start $\{2\}$ wins for large N. Both singleton atoms carry the factor $(1 - T/N)^{N-1}$, so $M_{\{1\}}/M_{\{2\}} = (\mu_1/\mu_2)e^{-T} = (\mu_1/\mu_2)N^{-1/4}$ exactly. For the start $\mu_2 = 1/(1 + 2N^{1/4})$ this ratio is 2 at every N. The law is linear in the start, so Lemma 2.4 for the two starts at a single state gives $M_{\{1,2\}} \le CLN^{-1/2}M_{\{1\}}$ with C free of μ. So $\{1\}$ wins at every large N, against the limiting prediction. The switch between $\{1\}$ and $\{2\}$ in the variable v_2 sits at $\log(\mu_1/\mu_2)/(\tau L) = 1 + 4\log 2/L$ instead of at $q_2 - q_1 = 0$. On $\mathcal{M}$ the same shift is at most $\log(1/\mu_*)/(\tau L)$, uniformly.

Remark 5.9 (the size of the correction). On the two-state class with $R = 3$ and $b = 1$, the chain $q_1 = q_2 = 1$ with the uniform start has $t_{12} = 6.16$ and $c_{12} = -16.5$, and at $N = 10^7$ the exact law gives $t_{12,N} = 4.65$, against 5.63 for $t_{12}(1 - \varepsilon_N)$ and 4.60 for the prediction of (b). So the term c_{ij}/L, which no correction fixed in advance removes, dominates $t_{ij}\varepsilon_N$ at accessible N.

6. Low temperature and the cycle hierarchy

We use the cycles of Freidlin and Wentzell [13] in the form of Catoni [6]. The depth of a face (Theorem 6.1) is a short consequence of Catoni's theorem on the exit time of a domain, and the prefactors (Proposition 6.3) are classical in substance [24]. Section 6.2 proves Theorem D. Its main steps are the depth of a face under an added calibrated killing (Lemma 6.9), the strict growth of depths along nested cycles (Lemma 6.10), and a comparison of the rewarded cycle with each kind of face.

6.1. Arrhenius families

Let $V_{ij} \geq 0$ for $i \neq j$ be fixed costs and let

$$q_{ij}(\eta) = e^{-V_{ij}/\eta}, \qquad \eta > 0,$$

so every off-diagonal rate is positive at every positive temperature. We write Q_η for the generator, and for a face F we write $\lambda_F(\eta)$ for the exit rate λ_F of F under Q_η. In this section the letter h is a height. For a proper face F define its depth

$$D(F) = \lim_{\eta \downarrow 0} \big(-\eta \log \lambda_F(\eta)\big),$$

whose existence is part of Theorem 6.1. Put $D(V) = \infty$. A fixed height $h > 0$ sets the time scale

$$\tau_\eta = e^{h/\eta}, \qquad T = \tau_\eta L.$$

Throughout the section we take $N \to \infty$ first at fixed η and then $\eta \downarrow 0$, and every threshold in N may depend on η. The start μ_η may depend on η but has full support and does not depend on N. Since all rates are positive, every face is irreducible, and with a full-support start every face satisfies (H) by Lemma 2.1(a). So Theorem 2.5(b) applies at every fixed η to every face. At fixed η the face maxima of Theorem 2.8 give

$$\log M_F = -\Phi_{\eta,h}(F)\, L + \frac{|F| - 1}{2} \log L + O_\eta(1), \qquad \Phi_{\eta,h}(F) = |F| - 1 + e^{h/\eta} \lambda_F(\eta). \tag{7}$$

So $\Phi_{\eta,h}$ is the exponent Φ_τ of Theorem 2.5 for the generator Q_η at $\tau = e^{h/\eta}$. The constants in (7) may depend badly on η, and nothing below exchanges the two limits.

We use the cycles of Freidlin and Wentzell in Catoni's convention [6, Definition 4.2]. A nonempty proper set C is a cycle if it is a singleton or if, for all distinct $x, y \in C$, the probability of leaving C before hitting y, starting from x, is exponentially small in $1/\eta$. Its depth $\Gamma(C)$ is the exponential scale of its mean exit time, which does not depend on the starting point in C [6, Proposition 4.6], and $\Gamma(\{i\}) = \min_{j \neq i} V_{ij}$. For a reversible family write $V_{ij} = H_{ij} - U_i$ with H symmetric, so that $\pi_i(\eta) \propto e^{-U_i/\eta}$. Let $b(C) = \min_{i \in C, j \notin C} H_{ij}$ be the boundary height of a proper set C with at least 2 states. Then C is a cycle exactly when any 2 of its points are joined by a path whose saddles H are all less than $b(C)$, and then $\Gamma(C) = b(C) - \min_{i \in C} U_i$ [8, Definition 3.5, Propositions 3.10 and 3.15]. We call this the energy criterion. The inequality is strict, since a weak one would admit sets at a common saddle level that are not cycles.

Theorem 6.1 (the depth of a face)**.** *For every nonempty proper face F the depth $D(F)$ exists and*

$$D(F) = \max\{\Gamma(C) : C \subseteq F \text{ a cycle}\}.$$

In particular $D(C) = \Gamma(C)$ for every cycle C, and F contains a cycle C with $D(C) = D(F)$. Moreover, with $\mathcal{G}_F = (-Q_F)^{-1}$ and $M_F^{\text{exit}} = \max_{i \in F} \mathbb{E}_i \tau_{F^c}$,

$$\frac{1}{M_F^{\text{exit}}} \leq \lambda_F(\eta) \leq \frac{|F|}{M_F^{\text{exit}}},$$

so that $D(F) = \lim_{\eta \downarrow 0} \eta \log M_F^{\text{exit}}$.

Proof. Fix η and a proper face F, and write λ_F for $\lambda_F(\eta)$. By Lemma 2.2 with $v = 0$, $\lambda_F > \lambda_V = 0$. So $\mathcal{G}_F = \int_0^\infty e^{tQ_F}\,dt$ converges, and since $(e^{tQ_F})_{ij} = \mathbb{P}_i(X_t = j,\ t < \tau_{F^c})$ for the chain X with generator Q_η,

$$(\mathcal{G}_F)_{ij} = \mathbb{E}_i \int_0^{\tau_{F^c}} \mathbf{1}\{X_t = j\}\,dt = \mathbb{P}_i(\tau_j < \tau_{F^c})\,(\mathcal{G}_F)_{jj} \le (\mathcal{G}_F)_{jj}$$

by the strong Markov property at τ_j.

The exit-time bounds. The row sums of $\mathcal{G}_F$ are the mean exit times $\mathbb{E}_i\tau_{F^c}$. The spectral radius ρ of a nonnegative matrix is at most its largest row sum and at least each of its diagonal entries. Hence

$$\rho(\mathcal{G}_F) \le M_F^{\text{exit}} \le \operatorname{tr}\mathcal{G}_F \le |F| \max_j (\mathcal{G}_F)_{jj} \le |F|\rho(\mathcal{G}_F).$$

The Perron vector of $-Q_F$ is positive and is an eigenvector of $\mathcal{G}_F$ for $1/\lambda_F$. A positive eigenvector of a nonnegative matrix belongs to its spectral radius [2], so $\rho(\mathcal{G}_F) = 1/\lambda_F$, which gives the bounds.

The depth. Fix $\varkappa > d$ and put $P_\eta = I + Q_\eta/\varkappa$. Every rate is at most 1, so every exit rate is at most $d-1$, and P_η is stochastic with holding probabilities at least $1 - (d-1)/\varkappa > 0$. Then $-\eta \log P_\eta(i,j) \to V_{ij}$ for $i \ne j$ and $-\eta \log P_\eta(i,i) \to 0$. So (P_η) has rare transitions in the sense of [6, Definition 2.1], with $\beta = 1/\eta$ and the rate function equal to V_{ij} off the diagonal and to 0 on it, and this rate function is irreducible since all rates are positive. The chain P_η visits the same states in the same order as X, so the two chains have the same cycles. The continuous chain is the discrete one run at the jump times of a Poisson clock of rate $\varkappa$, so its mean exit times are the discrete ones divided by $\varkappa$. For a domain $\mathcal{O}$ and $x \in \mathcal{O}$, Catoni's Proposition 4.2 [6] gives the limit $H_{\mathcal{O}}(x)$ of β^{-1} log of the mean exit time from $\mathcal{O}$ started at x. Let $\mathcal{M}(\mathcal{O})$ be the set of maximal cycles contained in $\mathcal{O}$ (his Definition 4.3), and $\Gamma(C)$ the depth of his Proposition 4.6. His Proposition 4.11 states that $\max_{x\in\mathcal{O}} H_{\mathcal{O}}(x) = \max\{\Gamma(C) : C \in \mathcal{M}(\mathcal{O})\}$ for every domain $\mathcal{O}$ other than the whole space. With $\mathcal{O} = F$ this reads

$$\lim_{\eta\downarrow 0} \eta \log M_F^{\text{exit}} = \max_{C\in\mathcal{M}(F)} \Gamma(C).$$

These are cycles of the whole chain, not of a chain restricted to F. By the exit-time bounds $-\eta\log\lambda_F$ differs from $\eta \log M_F^{\text{exit}}$ by at most $\eta \log|F|$, so $D(F)$ exists and equals this limit. Every cycle $C \subseteq F$ lies in some $C' \in \mathcal{M}(F)$, and $\Gamma(C) \le \Gamma(C')$ since $\tau_{C^c} \le \tau_{C'^c}$ pathwise. So the maximum over $\mathcal{M}(F)$ is the maximum over all cycles in F, attained at some $C \in \mathcal{M}(F)$. If F is a cycle, then $\mathcal{M}(F) = \{F\}$ and $D(F) = \Gamma(F)$. □

Theorem 6.2 (noncritical heights)**.** *Let $\mathcal{D}$ be the finite set of depths of proper cycles and let $h \notin \mathcal{D}$. Let $k(h)$ be the least size of a cycle of depth greater than h, where V counts as a cycle of infinite depth and size d. Then for all sufficiently small η, every minimizer of $\Phi_{\eta,h}$ is a cycle of size $k(h)$ and depth greater than h, and every eventual support of the global mode, in the limit $N \to \infty$ at that η, is such a cycle.*

Proof. By Theorem 6.1 the depth of every proper face lies in $\mathcal{D}$ and differs from h. If $D(F) < h$, then $e^{h/\eta}\lambda_F(\eta) = e^{(h-D(F)+o(1))/\eta} \to \infty$. If $D(F) > h$, then $e^{h/\eta}\lambda_F(\eta) \to 0$. The face V has the score $d-1$. Call a face deep if it is V or a proper face of depth greater than h.

A deep face of least size is a cycle. Indeed, a deep proper face F contains a cycle C with $D(C) = D(F) > h$ by Theorem 6.1, and C is deep, so $C = F$ if F has least size. Every cycle of depth greater than h is deep. So the deep faces of least size are the cycles of size $k(h)$ and depth greater than h.

There are finitely many faces, so for all sufficiently small η every face that is not deep has $\Phi_{\eta,h} > d$, and every deep face of size $k(h)$ has $\Phi_{\eta,h} < k(h)$. A deep face of larger size has $\Phi_{\eta,h} \geq k(h)$, since $\lambda_F \geq 0$. Hence every minimizer is a cycle of size $k(h)$ and depth greater than h, and Theorem 2.5(b) at $\tau = e^{h/\eta}$ puts the support of every global mode among the minimizers for large N. □

Theorem 6.2 is a theorem about least cardinality. It does not say that every cycle of size $k(h)$ and depth greater than h wins, and the cycles that are minimal for inclusion among the deep ones can have other sizes and are then never selected. Among faces of equal size, minimizing $\Phi_{\eta,h}$ is minimizing $\lambda_F(\eta)$ at fixed η, whatever h is. So the deeper cycle wins first, and at equal depth the smaller prefactor of Proposition 6.3. An exponentially small difference of rates still decides, because $N \to \infty$ first. If two scores agree exactly at some η, the terms of (7) below L may decide, but they need not. In the symmetric case, for instance two disjoint cycles built from the same costs with a start that is symmetric between them, the two faces have the same exit rate and the same constant, and nothing at any order separates them. The theorem then says only that the mode lies on one of the tied cycles, and Section 6.2 treats such ties as unobserved. The full-support start is needed, since otherwise some cycles are not admissible faces.

Proposition 6.3 (prefactors)**.** *For every proper face F there is $a_F \in (0, \infty)$ with $\lambda_F(\eta) = a_F e^{-D(F)/\eta}(1 + o(1))$, namely $a_F = 1/\rho(B_F)$ with $B_F = \lim_{\eta \downarrow 0} e^{-D(F)/\eta}\mathcal{G}_F(\eta)$, and*

$$D(F) = \nu\big(\det(-Q_F)\big) - \min_{i \in F} \nu\big(\det(-Q_{F\setminus\{i\}})\big),$$

where $\nu(f)$ is the exponent b of the first nonzero term $c\,e^{-b/\eta}$ of a finite sum of exponentials f, and the determinant of the empty matrix is 1.

Proof. Every minor of $-Q_F$ is a finite real combination of exponentials $e^{-b/\eta}$. After equal exponents are collected, a combination that is not identically zero has a first nonzero term $c\,e^{-\nu/\eta}$ and equals $c\,e^{-\nu/\eta}(1 + o(1))$. The determinant of $-Q_F$ is positive, since its real eigenvalues are at least $\lambda_F > 0$ and the others come in conjugate pairs. The entries of $\mathcal{G}_F$ are positive by the identity in the proof of Theorem 6.1. By Cramer's rule each is a cofactor divided by $\det(-Q_F)$, so

$$(\mathcal{G}_F)_{ij} = \beta_{ij} e^{e_{ij}/\eta}(1 + o(1)), \qquad \beta_{ij} > 0.$$

The diagonal cofactor of j is $\det(-Q_{F\setminus\{j\}})$, so $e_{jj} = \nu(\det(-Q_F)) - \nu(\det(-Q_{F\setminus\{j\}}))$. The exit-time bounds give $\max_j (\mathcal{G}_F)_{jj} \leq 1/\lambda_F \leq |F| \max_j (\mathcal{G}_F)_{jj}$, so $D(F) = \max_j e_{jj}$, which is the determinant formula. Since $(\mathcal{G}_F)_{ij} \leq (\mathcal{G}_F)_{jj}$, we have $e_{ij} \leq D(F)$ for all i, j. Hence $e^{-D(F)/\eta}\mathcal{G}_F$ converges entrywise to the matrix B_F with entries β_{ij} where $e_{ij} = D(F)$ and 0 elsewhere. It has a positive diagonal entry, so $\rho(B_F) > 0$. The spectral radius is continuous in the entries, and $\rho(e^{-D(F)/\eta}\mathcal{G}_F) = e^{-D(F)/\eta}/\lambda_F$. Hence $e^{D(F)/\eta}\lambda_F \to 1/\rho(B_F)$. □

The asymptotics of principal eigenvalues of matrices with entries of order $e^{-V_{ij}/\eta}$, and their expressions through graphs and determinants, are due to Wentzell [24] and Freidlin and Wentzell [13, Chapter 6], and the proof above is a short version for killed generators. The proposition uses the exact form of the rates. A family that satisfies only the logarithmic limits $-\eta \log q_{ij} \to V_{ij}$ need not have finite prefactors.

Theorem 6.4 (critical windows)**.** *Fix a depth $\gamma \in \mathcal{D}$ and $x > 0$, and let $T = xe^{\gamma/\eta}L$. Define the limiting score*

$$\mathcal{J}_{\gamma,x}(F) = \begin{cases} +\infty, & D(F) < \gamma, \\ |F| - 1 + xa_F, & D(F) = \gamma, \\ |F| - 1, & D(F) > \gamma, \end{cases}$$

with V in the last case. If $\mathcal{J}_{\gamma,x}$ has a unique minimizer F_x, then for all sufficiently small η, F_x is the unique minimizer of the score $|F| - 1 + xe^{\gamma/\eta}\lambda_F(\eta)$ and the eventual support of the global mode as $N \to \infty$ at that η. The same holds, with one threshold for η, uniformly for x in a compact interval on which one face has a positive gap. The lower envelope $x \mapsto \min_F \mathcal{J}_{\gamma,x}(F)$ is concave and piecewise linear, and its minimizers need not be cycles.

Proof. Let $S_\eta(F) = |F| - 1 + xe^{\gamma/\eta}\lambda_F(\eta)$, which is $\Phi_\tau(F)$ for Q_η at $\tau = xe^{\gamma/\eta}$. By Proposition 6.3, $e^{\gamma/\eta}\lambda_F(\eta) = a_F e^{(\gamma - D(F))/\eta}(1 + o(1))$, which tends to ∞, a_F or 0 as $D(F) < \gamma$, $D(F) = \gamma$ or $D(F) > \gamma$, and $S_\eta(V) = d - 1$. On a compact interval $I \subset (0, \infty)$ the convergence $S_\eta(F) \to \mathcal{J}_{\gamma,x}(F)$ is uniform in x, divergence included, since x is bounded below on I.

Suppose that one face F satisfies $\mathcal{J}_{\gamma,x}(G) \geq \mathcal{J}_{\gamma,x}(F) + g$ for all $G \neq F$ and $x \in I$, with $g > 0$. There are finitely many faces, so there is η_0 with $S_\eta(G) > S_\eta(F)$ for all $G \neq F$, $x \in I$ and $\eta < \eta_0$. A unique minimizer F_x at a single x has such a gap on a small interval around x, because the scores are continuous in x. Theorem 2.5(b) at $\tau = xe^{\gamma/\eta}$ then gives the support of the global mode, with a threshold in N that may depend on η and x.

The function $x \mapsto \mathcal{J}_{\gamma,x}(F)$ is affine for every face with $D(F) \geq \gamma$, and $\mathcal{J}_{\gamma,x}(V) = d - 1$ is finite. So the envelope is the minimum of finitely many affine functions, which is concave and piecewise linear. Example 6.5 shows a minimizer that is not a cycle. □

At a tie of the limiting lines, Theorem 6.4 identifies the candidates only, and the exact scores at each η decide. Along the envelope the slope a_F of the active line decreases with x, and its intercept $|F| - 1$ increases. The convex hull of first-order winners in [22, Theorem 4.4] behaves in the same way. Putting $h = \gamma$ and treating the faces of depth γ as faces of greater depth would discard the term xa_F, which is of order 1 in the coefficient of L.

Example 6.5 (a noncycle winner on an open window)**.** Take $U = (0, 1, 1, 1)$ and the symmetric saddles $H_{12} = 2$, $H_{13} = H_{23} = H_{34} = 3$, $H_{14} = H_{24} = 5$, so that with $t = e^{-1/\eta}$

$$(q_{ij})_{i \neq j} = \begin{pmatrix} * & t^2 & t^3 & t^5 \\ t & * & t^2 & t^4 \\ t^2 & t^2 & * & t^2 \\ t^4 & t^4 & t^2 & * \end{pmatrix}.$$

The only proper cycle with more than one state is $\{1, 2\}$, of depth 3. The faces of depth 3 are $\{1, 2\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$, with prefactors $a_{12} = 2$, $a_{123} = \frac{2}{3}$ and $a_{124} = 2$, so at $T = xt^{-3}L$

the limiting scores are $1+2x$, $2+\frac{2}{3}x$, $2+2x$ and 3 for V. Hence the eventual winner is $\{1,2\}$ for $0<x<\frac{3}{4}$, the noncycle $\{1,2,3\}$ for $\frac{3}{4}<x<\frac{3}{2}$, and V for $x>\frac{3}{2}$. At $h=3$ and $x=1$ the noncycle beats every cycle. Its window has width $\log 2$ in the rescaled height $(h-3)/\eta$ and collapses onto the single height $h=3$, which is why it does not contradict Theorem 6.2.

Proof. Here $q_{ij}=t^{H_{ij}-U_i}$ and $\pi\propto(1,t,t,t)$. The exit rates are

$$q_1=t^2+t^3+t^5, \qquad q_2=t+t^2+t^4, \qquad q_3=3t^2, \qquad q_4=t^2+2t^4.$$

Cycles and depths. The singletons have depths $\min_j V_{ij}=2,1,2,2$. We test the sets with 2 or 3 states by the energy criterion. The set $\{1,2\}$ has boundary height 3 and internal saddle $H_{12}=2<3$, so it is a cycle of depth $3-U_1=3$. A set that contains exactly one of the states 1 and 2 has the edge $H_{12}=2$ to its complement, and no two of its points communicate below 2. The remaining sets $\{3,4\}$, $\{1,2,3\}$ and $\{1,2,4\}$ have boundary height 3. In the first two the state 3 is joined to the others only by saddles of height 3, and in the third the state 4 is joined to the others only by saddles of height 5. So none is a cycle. By Theorem 6.1, a face that contains $\{1,2\}$ has depth 3 and every other proper face has depth at most 2. So $\mathcal{D}=\{1,2,3\}$, and the faces of depth 3 are the three named.

Prefactors. We apply Proposition 6.3. For $\{1,2\}$, $\det(-Q_{12})=q_1q_2-t^3=2t^4+O(t^5)$, and the first column of the adjugate is $(q_2,t)=(t,t)+O(t^2)$, while the second column is $(t^2,q_1)=O(t^2)$. Hence $t^3\mathcal{G}_{12}\to B_{12}$ with first column $(\frac{1}{2},\frac{1}{2})$ and second column 0, so $\rho(B_{12})=\frac{1}{2}$ and $a_{12}=2$. For $\{1,2,3\}$,

$$-Q_{123}=\begin{pmatrix} t^2+t^3+t^5 & -t^2 & -t^3 \\ -t & t+t^2+t^4 & -t^2 \\ -t^2 & -t^2 & 3t^2 \end{pmatrix}.$$

Its determinant is $2t^6+O(t^7)$. The first column of its adjugate has the leading terms $3t^3, 3t^3, 2t^3$, and every other entry of the adjugate is $O(t^4)$. So B_{123} has first column $(\frac{3}{2},\frac{3}{2},1)$ and zeros elsewhere. Its eigenvalues are $\frac{3}{2}$ and 0, so $a_{123}=\frac{2}{3}$. For $\{1,2,4\}$ the determinant is $2t^6+O(t^7)$, the first column of the adjugate is $(t^3,t^3,0)+O(t^4)$ and the other entries are $O(t^4)$. So B_{124} has first column $(\frac{1}{2},\frac{1}{2},0)$ and $a_{124}=2$. These leading terms are the output of an exact computation in rational arithmetic.

The winners. At $\gamma=3$ the faces of depth less than 3 have $\mathcal{J}_{3,x}=\infty$. The lines are $1+2x$, $2+\frac{2}{3}x$, $2+2x$ and 3. The face $\{1,2,4\}$ is always 1 above $\{1,2\}$. Then $1+2x<2+\frac{2}{3}x$ exactly when $x<\frac{3}{4}$, $2+\frac{2}{3}x<3$ exactly when $x<\frac{3}{2}$, and $1+2x<3$ exactly when $x<1$. So the minimizer is unique and equals $\{1,2\}$ on $(0,\frac{3}{4})$, $\{1,2,3\}$ on $(\frac{3}{4},\frac{3}{2})$ and V on $(\frac{3}{2},\infty)$, and Theorem 6.4 turns each into the eventual support of the global mode, uniformly on compact subintervals. We claim nothing at the ties $x=\frac{3}{4}$ and $x=\frac{3}{2}$.

The noncycle. At $x=1$, that is at $h=3$, the score of $\{1,2,3\}$ is $\frac{8}{3}$. The cycles of depth less than 3 have infinite limiting score, and the only cycles of depth at least 3 are $\{1,2\}$ and V, both of score 3. So the noncycle beats every cycle. In the rescaled height $s=(h-3)/\eta=\log x$ the window is $\log\frac{3}{4}<s<\log\frac{3}{2}$, of width $\log 2$. Its heights $h=3+\eta s$ tend to 3, so at every fixed $h\neq 3$ the window is eventually left. □

The mechanism is that state 3 costs one dimension but returns the chain to $\{1,2\}$ before it is killed at state 4, which lowers the leading exit rate from $2t^3$ to $\frac{2}{3}t^3$. On the middle

interval this gain is worth more than the dimension. The selected face is the support of the counts, not of the limiting occupation fractions. For $\{1,2,3\}$ the right Perron vector tends to $(1,1,\frac{2}{3})$ up to scale, and with $\pi \propto (1,t,t,t)$ the quasi-ergodic law α^*_{123} tends to the point mass at state 1.

Example 6.6 (a branch switch)**.** Take $U = (2,0,0)$, $H_{12} = H_{13} = 5$ and $H_{23} = 1$. Then $\{2,3\}$ is a cycle of depth 5, the singleton depths are $3,1,1$, and the winners are $\{1\}$ for $0 < h < 3$, $\{2,3\}$ for $3 < h < 5$ and V for $h > 5$. The two winning cycles are disjoint nodes of the same cycle tree, so the selector jumps between branches and never selects the nodes $\{2\}$ and $\{3\}$.

Proof. With $t = e^{-1/\eta}$ the rates are $q_{12} = q_{13} = t^3$, $q_{21} = q_{31} = t^5$ and $q_{23} = q_{32} = t$. So $\lambda_{\{1\}} = 2t^3$, $\lambda_{\{2\}} = \lambda_{\{3\}} = t + t^5$ and $\lambda_{\{2,3\}} = t^5$. By the energy criterion $\{2,3\}$ is a cycle, and $\{1,2\}$, $\{1,3\}$ are not, since each has an edge of height 1 to its complement. Hence $\mathcal{D} = \{1,3,5\}$. Theorem 6.2 gives the winner $\{1\}$ on $(0,3)\setminus\{1\}$, since $2t^3 < t + t^5$, the winner $\{2,3\}$ on $(3,5)$, where $k(h) = 2$, and V above 5. At $h = 1$, $\Phi_{\eta,1}(\{1\}) = 2t^2 \to 0$ while every other face has a score that tends to at least 1, so Theorem 2.5(b) applies directly. We claim nothing at $h = 3$ and $h = 5$. □

6.2. Reading the cycle tree

We observe only the winning face, as a function of the height and of a field, and ask how much of the cycle structure it shows. A field is a vector $v = v(\eta) \in \mathbb{R}^d$, which may depend on η but not on N. As in Section 4.1 we write $\lambda_F(v) = -\varrho(Q_{\eta,F} + \operatorname{diag} v_F)$, with $Q_{\eta,F}$ the killed generator of F under Q_η, and

$$\Phi_{\eta,h}(F;v) = |F| - 1 + e^{h/\eta}\lambda_F(v),$$

so that $\Phi_{\eta,h}(F;0) = \Phi_{\eta,h}(F)$. In this subsection a height h may be any real number. We say that a face G *wins at* (h,v) if there is $\eta_0 > 0$ such that for every $\eta < \eta_0$ the face G is the unique minimizer of $\Phi_{\eta,h}(\cdot\,;v)$ over all nonempty faces. Where two faces of equal size have identical rates, for instance two disjoint cycles built from the same costs, no face wins and nothing is observed.

Proposition 4.1 turns a win into an observation. At a fixed η it applies to Q_η, μ_η, the vector $v(\eta)$ of either sign and $\tau = e^{h/\eta}$. So if G wins at (h,v), then for every $\eta < \eta_0$ the support of every maximizer of the weighted law is G for all large N. Nothing else below is probabilistic.

Two elementary facts are used throughout. The Perron root of a Metzler matrix is nondecreasing in its entries [2, Ch. 2], so $v \le w$ entrywise gives $\lambda_F(w) \le \lambda_F(v)$, and $\lambda_F(v + \kappa\mathbf{1}) = \lambda_F(v) - \kappa$ for every real κ. So adding a constant to the field changes no winner. And by Lemma 2.2, $\lambda_G(v) < \lambda_F(v)$ whenever $F \subsetneq G$. In particular $D(F) \le D(G)$ for $F \subseteq G$.

Proposition 6.7 (the unperturbed path)**.** *Let $v = 0$, and let a proper face W win at some height $h_0 \notin \mathcal{D}$. Then W is a cycle with $D(W) > h_0$, W wins at every $h \in [h_0, D(W)) \setminus \mathcal{D}$, and W wins at no $h > D(W)$. Hence the supremum of the heights at which W wins is $D(W)$.*

Proof. By Theorem 6.2, W is a cycle of size $k(h_0)$ and depth greater than h_0. Let $h \in [h_0, D(W)) \setminus \mathcal{D}$. A face of depth greater than h has depth greater than h_0, so $k(h) \ge$

$k(h_0) = |W|$, and $k(h) = |W|$ since $D(W) > h$. By Theorem 6.2 again, for small η every minimizer of $\Phi_{\eta,h}$ is a cycle G with $|G| = |W|$ and $D(G) > h_0$. For such $G \neq W$, $\Phi_{\eta,h}(G) - \Phi_{\eta,h}(W) = e^{h/\eta}(\lambda_G - \lambda_W)$ has the sign of $\Phi_{\eta,h_0}(G) - \Phi_{\eta,h_0}(W)$, which is positive for small η. Hence W is the unique minimizer. For $h > D(W)$ we have $e^{h/\eta}\lambda_W \to \infty$, while $\Phi_{\eta,h}(V) = d - 1$. □

So the path shows the depth of each face it selects, as the supremum of the heights at which it wins, and nothing about the other faces. The heights at which a face wins may skip depths of faces that never win, and a noncycle winner of a critical window occupies a single height. In Example 6.5 the singleton $\{4\}$ wins on $(0,1) \cup (1,2)$, since $q_4 < q_1 < q_3$ and $q_4 < q_2$ for small η, then $\{1,2\}$ wins on $(2,3)$ and V above 3. The cut at 1 is the depth of the cycle $\{2\}$, which never wins. The path shows $D(\{4\}) = 2$ and $D(\{1,2\}) = 3$ and nothing about the cycles $\{1\}$, $\{2\}$ and $\{3\}$. The reference is absolute, since $\lambda_V(0) = 0$ makes the score of V exactly $d - 1$.

Killing the complement of a face at a large rate does not read its depth. The killing gives V an exit rate below λ_F, and the winner leaves F only at the height where the gap between λ_F and that rate, the flux that returns to F through the killed states, becomes visible. This gap can be of smaller order than λ_F. For two states, $F = \{1\}$ and the killing rate $e^{g/\eta}$ at state 2, the exit rate λ of V is the smaller root of $(q_{12} - \lambda)(q_{21} + e^{g/\eta} - \lambda) = q_{12}q_{21}$, so the gap is $q_{12}q_{21}/(q_{21} + e^{g/\eta} - \lambda) = e^{-(V_{12}+V_{21}+g)/\eta}(1 + o(1))$, and the winner leaves $\{1\}$ at the height $V_{12} + V_{21} + g$ and not at $D(\{1\}) = V_{12}$.

The cure is to keep a face whose rate is known. For the unperturbed path this is V, with $\lambda_V(0) = 0$. Every field $v \geq 0$ keeps a version of it, since then $\lambda_V(v) \leq 0$ and $\Phi_{\eta,h}(V; v) \leq d-1$. We therefore reward the target face instead of killing its complement. For a proper face G, a fixed constant $c > 0$ and a real γ put

$$v^{G,\gamma} = c\, e^{-\gamma/\eta} \mathbf{1}_G.$$

The field is constant on G, so $\lambda_G(v^{G,\gamma}) = \lambda_G - c\, e^{-\gamma/\eta}$ exactly. Define the threshold of G as

$$\vartheta(G) = \sup\big\{\gamma \in \mathbb{R} :\ G \text{ wins at } (h, v^{G,\gamma}) \text{ for some } h > \gamma\big\},$$

with $\sup \emptyset = -\infty$. It is read from the map $(h, v) \mapsto$ winner alone, at heights where a winner exists. The observer needs no list of critical heights, which depend on the depths.

The next 2 lemmas are standard facts of cycle theory. Laminarity is Catoni's Proposition 4.4 [6] and the tree of cycles of Trouvé [23] and of Olivieri and Vares [21, Chapter 6]. The strict growth of depths along nested cycles can be read off Catoni's Proposition 4.18. We include the short proofs for completeness, in the convention of cycles through hitting probabilities used here.

Lemma 6.8 (laminarity)**.** *Two cycles are disjoint or nested.*

Proof. Let C_1, C_2 be cycles with a common state z, and suppose that neither contains the other. Take $x \in C_1 \setminus C_2$ and $y \in C_2 \setminus C_1$. Since $y \notin C_1$, $\mathbb{P}_z(\tau_x < \tau_y) \geq \mathbb{P}_z(\tau_x < \tau_{C_1^c})$, which tends to 1 because C_1 is a cycle. Since $x \notin C_2$, $\mathbb{P}_z(\tau_{C_2^c} < \tau_y) \geq \mathbb{P}_z(\tau_x < \tau_y)$, which tends to 1 and contradicts that C_2 is a cycle. □

Lemma 6.9 (depth with a calibrated killing)**.** *Let $K \subseteq [d]$, $\gamma \in \mathbb{R}$ and $s = c\, e^{-\gamma/\eta}$. For every proper face G the limit $D^K(G) = \lim_{\eta \downarrow 0}\big(-\eta \log \lambda_G(-s\mathbf{1}_K)\big)$ exists, and*

$$D^K(G) = \max\{D^K(C) : C \subseteq G \text{ a cycle}\}.$$

Moreover $D^K(C) = D(C)$ if $C \cap K = \emptyset$, and $D^K(C) \leq \gamma$ if $C \subseteq K$.

Proof. Fix Λ larger than every element of $\mathcal{D}$. Add a cemetery state ∂. Every state of K jumps to ∂ at rate s, every other state of $[d]$ jumps to ∂ at rate $\varepsilon = e^{-\Lambda/\eta}$, and ∂ jumps to every state of $[d]$ at rate ε. All rates of this enlarged chain are positive. If $\gamma < 0$ we multiply all its rates by $e^{\gamma/\eta}$, which shifts every depth by $-\gamma$ and changes no hitting probability, hence no cycle. The proof of Theorem 6.1 uses the rates only through the limits of $-\eta \log$ of the rates and through the exit-time bounds, once the uniformization constant $\varkappa$ exceeds every exit rate. After the rescaling every rate between two states of $[d]$ is at most 1, every rate into ∂ is at most $\max(c, \varepsilon)$, and ∂ has the exit rate $d\varepsilon$. So every exit rate is at most $d - 1 + \max(c, 1)$, and $\varkappa > d + c$ serves. The bound covers the case where some $V_{ij} = 0$ and $\gamma = 0$, in which the rates 1 and c occur exactly. So the proof applies to the enlarged chain.

For $G \subseteq [d]$ the killed generator of G in the enlarged chain is $Q_G - \operatorname{diag}(s\mathbf{1}_{G\cap K} + \varepsilon \mathbf{1}_{G\setminus K})$. By the two elementary facts its exit rate λ'_G satisfies $\lambda_G(-s\mathbf{1}_K) \leq \lambda'_G \leq \lambda_G(-s\mathbf{1}_K) + \varepsilon$. Since $\lambda_G(-s\mathbf{1}_K) \geq \lambda_G$ and $D(G) < \Lambda$, we get $\lambda'_G = \lambda_G(-s\mathbf{1}_K)(1 + o(1))$. So Theorem 6.1 for the enlarged chain gives the existence of $D^K(G)$ and the identity $D^K(G) = \max D^K(C)$ over the cycles $C \subseteq G$ of the enlarged chain.

Each such C is a cycle of Q_η. Indeed, the enlarged chain is the chain X of Q_η run until an independent killing time, and then ∂. If X leaves C before it hits y, so does the enlarged chain. So for distinct $x, y \in C$ the probability $\mathbb{P}_x(\tau_{C^c} < \tau_y)$ for X is at most the same probability for the enlarged chain, which is exponentially small. Hence $D^K(G)$ is at most the maximum of $D^K(C)$ over the cycles $C \subseteq G$ of Q_η, and the reverse inequality holds by Lemma 2.2. If $C \cap K = \emptyset$ then $\lambda_C(-s\mathbf{1}_K) = \lambda_C$. If $C \subseteq K$ then $\lambda_C(-s\mathbf{1}_K) = \lambda_C + s \geq s$, so $D^K(C) \leq \gamma$. □

Lemma 6.10 (strict growth along nested cycles)**.** *If $C' \subsetneq C$ are cycles, then $D(C') < D(C)$. Hence $D(S) < D(C)$ for every nonempty proper subset S of a cycle C.*

Proof. Fix $z \in C \setminus C'$. Since C is a cycle there is $\alpha > 0$ with $\mathbb{P}_w(\tau_{C^c} < \tau_y) \leq e^{-\alpha/\eta}$ for all distinct $w, y \in C$ and all small η. For each η choose $y \in C'$ with $\mathbb{E}_y \tau_{C'^c} = M^{\text{exit}}_{C'}$. Start at y and cut the path into rounds. A round runs until the chain leaves C'. If it then sits at a state $w \in C \setminus C'$, the round continues until the chain returns to y or leaves C. A round ends outside C only if the chain leaves C' straight into C^c, which forces $\tau_{C^c} < \tau_z$, or if it leaves C from w before it reaches y. By the strong Markov property each round ends outside C with probability at most $2e^{-\alpha/\eta}$, and the rounds are independent and identically distributed. So the number R of rounds up to the exit from C has $\mathbb{E}R \geq e^{\alpha/\eta}/2$. Each round lasts at least its first part, of mean $M^{\text{exit}}_{C'}$, and R is a stopping time for the rounds. Wald's identity gives

$$M^{\text{exit}}_C \geq \mathbb{E}_y \tau_{C^c} \geq \mathbb{E}R \cdot M^{\text{exit}}_{C'} \geq \tfrac{1}{2} e^{\alpha/\eta} M^{\text{exit}}_{C'},$$

and $D(C) \geq D(C') + \alpha$ by the last statement of Theorem 6.1. For the second claim, Theorem 6.1 gives a cycle $C'' \subseteq S$ with $D(C'') = D(S)$, and $C'' \subsetneq C$. □

Theorem 6.11 (depths from a calibrated attraction)**.** *Fix $c > 0$.*

(a) Let G be a proper face. If G wins at $(h, v^{G,\gamma})$ for some $h > \gamma$, then $\gamma \le D(G)$. Hence $\vartheta(G) \le D(G)$.

(b) Let C be a cycle, and put $D^-(C) = \max_{i \in C} D(C \setminus \{i\})$, with $D^-(C) = -\infty$ for a singleton. Then $D^-(C) < D(C)$, and for every $\gamma < D(C)$ the cycle C wins at $(h, v^{C,\gamma})$ for every $h \in (\max(\gamma, D^-(C)), D(C))$. Hence $\vartheta(C) = D(C)$.

(c) For every proper face F,

$$D(F) = \max_{G \subseteq F} \vartheta(G).$$

Proof. Write $s = c\, e^{-\gamma/\eta}$ and $\sigma = e^{h/\eta}$, and $\Phi(G)$ for the score of G under the field.

(a) Let $\gamma > D(G)$ and $h > \gamma$. Then s is of smaller order than λ_G, so $\sigma(\lambda_G - s) \ge \sigma\lambda_G/2 \to \infty$. Since $v^{G,\gamma} \ge 0$, the score of V is at most $d - 1$, so G is not a minimizer for small η.

(b) The inequality $D^-(C) < D(C)$ is Lemma 6.10. Fix $h \in (\max(\gamma, D^-(C)), D(C))$ and write $v = v^{C,\gamma}$. The score of C is $|C| - 1 + \sigma\lambda_C - \sigma s$, where $\sigma\lambda_C \to 0$ because $h < D(C)$ and $\sigma s = c\, e^{(h-\gamma)/\eta} \to \infty$ because $h > \gamma$. On another face G the field is $s\mathbf{1}_{G\cap C} = s\mathbf{1}_G - s\mathbf{1}_{G\setminus C}$, so

$$\lambda_G(v) = \lambda_G(-s\mathbf{1}_{G\setminus C}) - s.$$

If $G \cap C = \emptyset$, then $\lambda_G(v) = \lambda_G > 0$, so $\Phi(G) \ge 0$ while $\Phi(C) \to -\infty$. *If $G \subsetneq C$*, then $\lambda_G(v) - \lambda_C(v) = \lambda_G - \lambda_C$, and $D(G) \le D^-(C) < h < D(C)$, so $\sigma(\lambda_G - \lambda_C) \ge \sigma\lambda_G/2 \to \infty$. *If $G \supsetneq C$*, then $\lambda_C(v) - \lambda_G(v) = \lambda_C - \lambda_G(-s\mathbf{1}_{G\setminus C}) \le \lambda_C$, so $\Phi(G) - \Phi(C) \ge |G| - |C| - \sigma\lambda_C \to |G| - |C| \ge 1$. The face V is of this kind. *Otherwise* G meets C without containing it or lying in it. Put $K = G \setminus C$. Then $\lambda_G(v) - \lambda_C(v) = \lambda_G(-s\mathbf{1}_K) - \lambda_C$. A cycle $C' \subseteq G$ that met both C and K would be nested with C by Lemma 6.8, which is impossible since C' meets K and $C \not\subseteq G$. So C' lies in $G \cap C$ or in K. In the first case Lemma 6.9 gives $D^K(C') = D(C') \le D(G \cap C) \le D^-(C)$, and in the second $D^K(C') \le \gamma$. By Lemma 6.9, $D^K(G) \le \max(D^-(C), \gamma) < h$, so $\sigma\lambda_G(-s\mathbf{1}_K) \to \infty$, while $\sigma\lambda_C \to 0$.

In every case $\Phi(G) - \Phi(C)$ tends to $+\infty$ or has limit inferior at least 1. There are finitely many faces, so C wins at (h, v). The interval of h is nonempty for every $\gamma < D(C)$, so $\vartheta(C) \ge D(C)$, and (a) gives equality.

(c) For $G \subseteq F$, (a) and the monotonicity of D give $\vartheta(G) \le D(G) \le D(F)$. By Theorem 6.1 there is a cycle $C \subseteq F$ with $D(C) = D(F)$, and $\vartheta(C) = D(F)$ by (b). □

The theorem needs no uniqueness. Under its own attraction the score of C tends to $-\infty$, so C wins even when other faces tie, as for two disjoint cycles of equal rates. The unperturbed path has no such property. Only the threshold in γ is used, since above $D(C)$ the faces that contain C compete through $\sigma(\lambda_C(v) - \lambda_G(v))$, which can diverge. A noncycle may win under its own attraction and its threshold may reach $D(G)$, which is why (c) takes the maximum over all subfaces and needs no prior knowledge of the cycles. In the deep-singleton family with $V_{12} = V_{21} = 1$, $V_{13} = V_{23} = 3$ and $V_{31} = V_{32} = 5$, the singleton $\{3\}$ of depth 5 wins on all of $(0, 5)$ along the unperturbed path, so the path never shows the cycle $\{1, 2\}$ of depth 3, while Theorem 6.11(b) reads $\vartheta(\{1, 2\}) = 3$.

Theorem 6.12 (cycles from depths)**.** *For a nonempty proper face C the following are equivalent.*

(i) C *is a cycle.*

(ii) $D(C') < D(C)$ *for every nonempty proper subset* C' *of* C*.*

(iii) $|C| = 1$*, or* $D(C \setminus \{i\}) < D(C)$ *for every* $i \in C$*.*

Moreover $\Gamma(C) = D(C)$ *for every cycle. Hence the depth function determines the set of cycles, their depths and the cycle tree, that is, the laminar family of cycles ordered by inclusion with root* V*, and depths increase strictly along every branch.*

Proof. (i) implies (ii) by Lemma 6.10, and (ii) and (iii) are equivalent since D is monotone. Suppose (ii). Theorem 6.1 gives a cycle $C^* \subseteq C$ with $D(C^*) = D(C)$, and (ii) forces $C^* = C$. The identity $\Gamma(C) = D(C)$ is in Theorem 6.1, laminarity is Lemma 6.8, and the strict growth is Lemma 6.10. □

No nondegeneracy is assumed. Ties between the depths of different faces are allowed, because the cycles are strict and Lemma 6.10 rules out a tie along nested cycles. The weak form of the criterion fails. Take 3 states with all $V_{ij} = 1$. Then $\lambda_{\{1\}} = 2e^{-1/\eta}$ and $\lambda_{\{1,2\}} = e^{-1/\eta}$, so $D(\{1\}) = D(\{2\}) = D(\{1,2\}) = 1$. The set $\{1,2\}$ is not a cycle, since from state 1 the chain leaves it before it hits 2 with probability $\frac{1}{2}$. Criterion (ii) rejects it, and the weak criterion $D(C') \le D(C)$ accepts it. In Example 6.5 the faces $\{1,2,3\}$ and $\{1,2,4\}$ tie in depth with the cycle $\{1,2\}$ and are rejected for the same reason. Theorems 6.11 and 6.12 together give Theorem D.

The prefactors are read at a critical depth from the unperturbed path. Fix $\gamma \in \mathcal{D}$, take $v = 0$ and $T = xe^{\gamma/\eta}L$, and let $\mathcal{J}_{\gamma,x}$ be the limiting score of Theorem 6.4. This is the height $h = \gamma + \eta \log x$, which moves with η, so here we observe the winner as a function of the window coordinate x. Write $e(x) = \min_F \mathcal{J}_{\gamma,x}(F)$ for the lower envelope, and put $\bar{a}_G = a_G$ if $D(G) = \gamma$ and $\bar{a}_G = 0$ if $D(G) > \gamma$, so that $\mathcal{J}_{\gamma,x}(G) = |G| - 1 + x\bar{a}_G$ when $D(G) \ge \gamma$.

Theorem 6.13 (prefactors from window endpoints)**.** *Suppose there are* $0 = x_0 < x_1 < \cdots < x_m < x_{m+1} = \infty$ *and faces* $G_0, \dots, G_m$ *with* $G_{k-1} \neq G_k$ *such that* G_k *is the unique minimizer of* $\mathcal{J}_{\gamma,x}$ *for every* $x \in (x_k, x_{k+1})$*.*

(a) $D(G_m) > \gamma$*, and* $D(G_k) = \gamma$ *for* $k < m$*.*

(b) $\bar{a}_{G_m} = 0$*, and for* $k = m, \dots, 1$*,*

$$\bar{a}_{G_{k-1}} = \bar{a}_{G_k} + \frac{|G_k| - |G_{k-1}|}{x_k}.$$

Hence the points x_k *and the sizes determine* a_{G_k} *for every* $k < m$*.*

(c) Let G *be a face of depth* γ *other than* $G_0, \dots, G_m$*, and put* $s_G = \sup_{x>0}(e(x) - |G| + 1)/x$*. Then* $a_G \ge s_G$*, and every value* $a > s_G$ *in place of* a_G *gives the same unique minimizers. If* $G' \subseteq G$ *has depth* γ*, then* $a_G \le a_{G'}$*.*

Proof. By Theorem 6.4, G_k is the eventual winner, uniformly on compact subintervals of (x_k, x_{k+1}), so the faces G_k and the points x_k are observed. The envelope is continuous, so the lines of G_{k-1} and G_k meet at x_k, that is, $|G_{k-1}| - 1 + x_k \bar{a}_{G_{k-1}} = |G_k| - 1 + x_k \bar{a}_{G_k}$. This is the recursion in (b). Since G_{k-1} lies below G_k to the left of x_k and above it to the right, $\bar{a}_{G_{k-1}} > \bar{a}_{G_k} \geq 0$, and a face with $\bar{a}_G > 0$ has depth γ. As $x \to \infty$ every line of positive slope diverges, while $\mathcal{J}_{\gamma,x}(V) = d - 1$. So the last piece has slope 0, that is, $D(G_m) > \gamma$ and $\bar{a}_{G_m} = 0$. This proves (a) and (b). For (c), the envelope is the minimum of the lines, so $|G| - 1 + x a_G \geq e(x)$ for all $x > 0$, which is $a_G \geq s_G$. If $a > s_G$, then $|G| - 1 + xa > e(x)$ for all $x > 0$, and the unique minimizers do not change. Finally $\lambda_G \leq \lambda_{G'}$ for $G' \subseteq G$, and at equal depths Proposition 6.3 turns this into $a_G \leq a_{G'}$. □

The reference is absolute because the last piece has slope exactly 0, and V, with $\lambda_V(0) = 0$, guarantees a face with a flat line. In Example 6.5 at $\gamma = 3$ the pieces are $G_0 = \{1, 2\}$, $G_1 = \{1, 2, 3\}$ and $G_2 = V$, with $x_1 = 3/4$, $x_2 = 3/2$ and sizes $2, 3, 4$. The recursion gives $a_{123} = (4-3)/(3/2) = 2/3$ and $a_{12} = 2/3 + (3-2)/(3/4) = 2$, the values found in the example. The face $\{1, 2, 4\}$ is never on the envelope. Here $s_{124} = 2/3$, attained on the whole interval $(3/4, 3/2)$, and $\{1, 2\} \subseteq \{1, 2, 4\}$ gives $a_{124} \leq a_{12} = 2$. So $2/3 \leq a_{124} \leq 2$. At $a_{124} = 2/3$ the line of $\{1, 2, 4\}$ would coincide with that of $\{1, 2, 3\}$ on the middle interval, so the hypothesis of Theorem 6.13 excludes the lower end, but the observed winners alone do not. The window thus recovers $a_{12} = 2$ and $a_{123} = 2/3$ and only bounds a_{124}, whose value is 2. A calibrated field need not help, because a field changes the rate of V to a value that depends on Q. In this example the limiting quasi-stationary laws of the 3 faces of depth 3 and the limit of π all sit at state 1. So a field $c\, e^{-3/\eta}$ at state 1 moves every line of the window by $-cx$ to first order, and a field of this order at another state moves no line.

Remark 6.14 (what the design needs). No step of this subsection uses reversibility. Theorem 6.1 rests on Catoni's theorem and the exit-time bounds, and the other lemmas use only the Green matrix, the definition of a cycle and the strong Markov property. Lemma 6.9 applies Theorem 6.1 to a chain with a cemetery, which is never reversible. The selection step is Proposition 4.1, which holds for a general generator.

7. Computations and open questions

Every statement is proved in the text, and no proof rests on a computation. The scripts below check parts of the paper by assertions, with a tolerance in floating point.

The script `verify_reconstruction.py` checks Sections 3 to 5. It verifies the inversion formula of Theorem 5.2(b) and the three-state identities of Remark 3.6 symbolically, and the family of Remark 5.4. It recovers Q from the strict winners on 40 random reversible chains with 4 states, with Lemma 5.1 checked on a grid of every design. It checks the constant $20/3 + 28/9 < 10$ in the proof of Theorem 5.5 and the bounds ϕ_* and g_* on 2000 chains. On 64 random generators with a rank-one cut it checks that the generator point of the cut-transpose has the principal minors of $-Q$ (in 49 of them it is neither Q nor Q^*), and it checks the four-state example after Theorem 3.5 in rational arithmetic. It checks the derivative used in the proof of Lemma 3.2, and on 200 generators with 3 states it checks that every generator with the same minors is Q or Q^*. The script `verify_finite_horizon.py` computes the entry times of Theorem 5.7 on a path with 3 states for $N = 10^3$ to 10^6 from the exact run expansion of [22,

Lemma 2.3], and the numbers of Remark 5.9. The script `verify_threshold.py` computes the least N after Theorem 5.7 for 2 chains and $b = 2^{-1}, \dots, 2^{-8}$, with Nb^2 and $Nb^2/\log N$. The script `verify_cycles.py` redoes Examples 6.5 and 6.6 in exact arithmetic. The script `verify_attraction.py`, with the module `attraction_core.py`, tests Theorem 6.11 on 11 cost matrices and 138 proper faces at $t = 10^{-32}$ and 10^{-48}, with the exact finite-η scores. Every cycle has the grid threshold $D(G) - 1/4$, no noncycle exceeds it, and $\max_{G \subseteq F} \vartheta(G)$ gives $D(F)$ on every face.

Some finite and algebraic steps are also proved in Lean 4 with Mathlib, in `lean/PaperC` of the same repository. They cover the multiaffine expansion of R_F and the normalization (N). They also cover the fact that a diagonal similarity keeping an irreducible Metzler matrix Metzler has entries of one sign, Theorem 3.5(d) for positive rates, and the determinants, adjugates and prefactors of Example 6.5. Mathlib has no Perron and Frobenius theorem for irreducible Metzler matrices, so the positive null vector in (N) is a hypothesis there.

Open questions. The crossing equation of [22, Theorem 5.8] for the weighted law would read the amplitudes of Proposition 3.8 off the modal-support map and separate the fiber of Theorem 3.5 by labels alone, but it is not proved. We do not know whether a single irreducible face is always traced against a singleton rather than a larger subface in the exposure lemma. At low temperature it is open whether some calibrated field separates a_{124} from a_{12} in Example 6.5, and a version of Theorem 6.11 with η tied to N is open.

Data and code availability

The code and data that support this paper are in the folder `kagey-problems/problem131/paperC2` of https://github.com/apemm/Kagey-Problems. The command `py -I verify_all.py`, run there with Python 3, `numpy`, `scipy`, `sympy` and `mpmath`, reruns the checks of Section 7. Each output goes to `paperC2/data`, headed by the sha256 of its script.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work grew out of Problem 131 of Peter Kagey's Open Problem Collection [17]. We thank him for posing it.

References

[1] A. Al Ahmadieh, The fiber of the principal minor map, preprint, arXiv:2309.00806v2, 2023. https://arxiv.org/abs/2309.00806v2.

[2] A. Berman, R. J. Plemmons, Nonnegative Matrices in the Mathematical Sciences, SIAM, Philadelphia, 1994. https://doi.org/10.1137/1.9781611971262.

[3] L. Bertini, A. Faggionato, D. Gabrielli, From level 2.5 to level 2 large deviations for continuous time Markov chains, Markov Process. Related Fields 20 (2014) 545–562. https://arxiv.org/abs/1212.6908.

[4] L. Bertini, A. Faggionato, D. Gabrielli, Large deviations of the empirical flow for continuous time Markov chains, Ann. Inst. Henri Poincaré Probab. Stat. 51 (2015) 867–900. https://doi.org/10.1214/14-AIHP601.

[5] L. Bertini, D. Gabrielli, C. Landim, Metastable Γ-expansion of finite state Markov chains level two large deviations rate functions, Ann. Appl. Probab. 34 (2024) 3820–3869. https://doi.org/10.1214/24-AAP2051.

[6] O. Catoni, Simulated annealing algorithms and Markov chains with rare transitions, in: Séminaire de Probabilités XXXIII, Lecture Notes in Math. 1709, Springer, Berlin, 1999, pp. 69–119. https://www.numdam.org/item/SPS_1999__33__69_0/.

[7] A. Chatterjee, S. Ghosh, R. Gurjar, R. Raj, Characterizing and testing principal minor equivalence of matrices, preprint, arXiv:2410.01961v1, 2024. https://arxiv.org/abs/2410.01961v1.

[8] E. N. M. Cirillo, F. R. Nardi, J. Sohier, Metastability for general dynamics with rare transitions: escape time and critical configurations, J. Stat. Phys. 161 (2015) 365–403. https://doi.org/10.1007/s10955-015-1334-6.

[9] J. E. Cohen, Convexity of the dominant eigenvalue of an essentially nonnegative matrix, Proc. Amer. Math. Soc. 81 (1981) 657–658. https://doi.org/10.1090/S0002-9939-1981-0601750-2.

[10] J.-F. Delmas, D. Dronnier, P.-A. Zitt, Transformations preserving the effective spectral radius of a matrix, preprint, arXiv:2406.12857v1, 2024. https://arxiv.org/abs/2406.12857v1.

[11] M. D. Donsker, S. R. S. Varadhan, On a variational formula for the principal eigenvalue for operators with maximum principle, Proc. Natl. Acad. Sci. USA 72 (1975) 780–783. https://doi.org/10.1073/pnas.72.3.780.

[12] G. M. Engel, H. Schneider, Matrices diagonally similar to a symmetric matrix, Linear Algebra Appl. 29 (1980) 131–138. https://doi.org/10.1016/0024-3795(80)90234-7.

[13] M. I. Freidlin, A. D. Wentzell, Random Perturbations of Dynamical Systems, third ed., Grundlehren Math. Wiss. 260, Springer, Heidelberg, 2012. https://doi.org/10.1007/978-3-642-25847-3.

[14] S. Friedland, Convex spectral functions, Linear Multilinear Algebra 9 (1981) 299–316. https://doi.org/10.1080/03081088108817381.

[15] I. J. Good, The frequency count of a Markov chain and the transition to continuous time, Ann. Math. Statist. 32 (1961) 41–48. https://doi.org/10.1214/aoms/1177705138.

[16] D. J. Hartfiel, R. Loewy, On matrices having equal corresponding principal minors, Linear Algebra Appl. 58 (1984) 147–167. https://doi.org/10.1016/0024-3795(84)90209-X.

[17] P. Kagey, Problem 131, Open Problem Collection. https://peterkagey.com/problems/131/, 2026 (accessed October 2026).

[18] S. Kim, C. Landim, Γ-expansion of the measure-current large deviations rate functional of non-reversible finite-state Markov chains, Stochastic Process. Appl. 189 (2025) 104733. https://doi.org/10.1016/j.spa.2025.104733. Section 5 is cited with the numbering of arXiv:2412.13515v2.

[19] C. Landim, Metastability from the large deviations point of view: a Γ-expansion of the level two large deviations rate functional of non-reversible finite-state Markov chains, Stochastic Process. Appl. 165 (2023) 275–315. https://doi.org/10.1016/j.spa.2023.09.001.

[20] R. Loewy, Principal minors and diagonal similarity of matrices, Linear Algebra Appl. 78 (1986) 23–64. https://doi.org/10.1016/0024-3795(86)90015-7.

[21] E. Olivieri, M. E. Vares, Large Deviations and Metastability, Encyclopedia Math. Appl. 100, Cambridge University Press, Cambridge, 2005. https://doi.org/10.1017/CBO9780511543272.

[22] A. Pemmasani, Occupation counts of rarely switching Markov chains: stratified local laws, support selection and the boundary layer, preprint, arXiv:2610.09258, 2026. https://arxiv.org/abs/2610.09258.

[23] A. Trouvé, Cycle decompositions and simulated annealing, SIAM J. Control Optim. 34 (1996) 966–986. https://doi.org/10.1137/S0363012993258586.

[24] A. D. Wentzell [Ventcel'], On the asymptotics of eigenvalues of matrices with elements of order $\exp\{-V_{ij}/(2\varepsilon^2)\}$, Dokl. Akad. Nauk SSSR 202 (1972) 263–265 (in Russian).

[25] G. Wolfer, S. Watanabe, Information geometry of reversible Markov chains, Inf. Geom. 4 (2021) 393–433. https://doi.org/10.1007/s41884-021-00061-6.